Which forks? The definition of a forking token can flip its relation to entropy

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Abstract

Token-level entropy is widely used to find the "forking tokens" of chain-of-thought reasoning: positions where a different choice would change the outcome. A fork can be measured by forcing an alternative token (a *forced swap*) or by *resampling* from the position. We measure both at the same positions, in a series of mostly pre-registered experiments with replay audits, on GSM-Symbolic arithmetic with three open models (OLMo-2 1B, Gemma-2-2b-it and Qwen2.5-7B-Instruct). On these models, entropy relates to the two definitions in opposite directions. Large forced-swap effects concentrate on confident, answer-bearing digit tokens (a substantial part of the pattern, not all of it), so entropy points away from them (within-item AUROC 0.20–0.42 in five experiments; confirmed under pre-registration in four). In Gemma this persisted when the swapped-in token's probability was held near 10^{-6}, so it is not only an effect of swapping in improbable tokens. Large resampling drops, estimated with a stratified estimator, align with entropy (0.75–0.89), but largely by construction: the drop factors exactly as $(1 - p_1)(q_{\text{alt}} - q_{\text{keep}})$, and keeping each position's $1 - p_1$ while shuffling the effects reproduces the alignment (post hoc, in the two smaller models). The same identity gives a lower-variance estimator of the drop. The two views meet at rare alternatives: in Gemma, an alternative the model itself samples at a confident position was about as damaging as a forced one, but such positions contributed almost nothing to how sampled answers vary. On a synthetic name-lookup task (Gemma, one hop) the anti-alignment did not appear. In the 7B model, six of seven pre-registered tests came out as predicted; the resampling side was only weakly measured there, and the test of rare positions was undecided. Studies that use entropy as a forking signal should state which definition they need and report answer-bearing tokens separately.

1 Introduction

High-entropy tokens have been proposed as the "forks" of reasoning: the minority of positions where the model hesitates, and where training signal or analysis should concentrate (Wang et al., 2025; Cheng et al., 2026). Direct tests of whether such positions change outcomes use two operationalisations:

- **Forced swap.** Replace the chosen token with an alternative (typically the runner-up) and check whether the answer changes (Ruan et al., 2025; at the sentence level, off-policy edits in Macar et al., 2026).
- **Resampling.** Sample the token and its continuation from a prefix and measure how the answer distribution moves (at the sentence level, Bogdan et al., 2025; Macar et al., 2026). Forking Paths (Bigelow et al., 2025) combines the two: it forces each probable alternative token, samples continuations after it, and weights the outcomes by the token's probability.

A forced swap asks *"what if this token were different?"*, ignoring how likely that is. Resampling asks *"does the outcome actually branch here?"*, which combines the effect of an alternative with the chance of sampling one. We show that on arithmetic this difference reverses the direction of entropy's relation to forks, and we take the reversal apart. One half of it is built into the resampling definition. The other half is the empirical finding: forced-swap effects concentrate on confident, answer-bearing digit tokens. That digits matter in arithmetic is not surprising; what matters for entropy-based methods is that, under one definition, selecting tokens by entropy systematically passes over them.

Findings.

1. **Forced-swap effects sit at confident digit tokens (§3.1, §3.3).** Entropy is anti-aligned with these effects in all five arithmetic experiments, and the anti-alignment was confirmed under pre-

registration in four, on three models (1.5B to 7.6B parameters, three families). Digit tokens carry a substantial part of it, not all. In Gemma it persisted when the probability of the swapped-in token was fixed at about 10^{-6} at every position, although the matched token at a digit position was often itself a digit. On a synthetic name-lookup task it did not appear (§3.4), so it may be specific to arithmetic.

2. **The resampling side is largely built in (§3.2).** The resampling drop factors exactly into the chance that an alternative is sampled, $1 - p_1$, times the effect of an alternative. A baseline that keeps $1 - p_1$ and shuffles the effects within items reproduces entropy's alignment with the drop (post hoc). That alignment is largely a property of the definition, not a finding about where reasoning branches.
3. **Rare alternatives (§3.5).** In Gemma, an alternative that the model itself samples was about as damaging as the forced runner-up, and most damaging where alternatives are rarest; but such positions contributed almost nothing to how the answer varies under sampling. Our reading (an interpretation, not a registered test) is that the two definitions look at the same positions from opposite sides.
4. **A larger model of a third family (§3.6).** In Qwen2.5-7B-Instruct the anti-alignment, the weighting, the digit concentration and the first two rarity results replicated under the original decision rules. The alignment with the resampling drop was confirmed under a new rule but only weakly measured, probably because that model leaves almost no drop at most positions. Whether rare positions contribute little was undecided, and the probability-matched swap was made descriptive before the run (a smoke run had found a token near 10^{-6} at only 47% of positions).

Contributions. (a) A same-position, within-item comparison of entropy against both fork definitions (five experiments, three model families), within a series of ten pre-registered experiments with frozen decision rules and replay audits; every registered test and its outcome, including negative results such as a transcoder study (Appendix C), is listed in Appendix B. (b) An exact decomposition that explains the resampling side and a weight-only baseline that shows how much of it is built in. (c) A probability-matched swap that separates "the model is confident here" from "the swapped-in token is improbable". (*d*) A same-position comparison of natural and forced alternatives across bands of rarity, which separates the size of an effect from its contribution to answer variation. (*e*) A practical tool: a stratified estimator of the resampling drop that makes per-position estimates usable at feasible sample sizes.

2 Setup

Models.

- OLMo-2 1B (`allenai/OLMo-2-0425-1B-Instruct`; Team OLMo et al., 2025; fixed revision, float16).
- Gemma-2-2b-it (`google/gemma-2-2b-it`; Gemma Team, 2024; fixed revision, bfloat16, eager attention).
- Qwen2.5-7B-Instruct (`Qwen/Qwen2.5-7B-Instruct`; Qwen Team, 2024; fixed revision, bfloat16, PyTorch's scaled-dot-product attention), used only in the replication DP9 (§3.6). Its own chat template, which inserts a default system message, is used unchanged.

Data. GSM-Symbolic (Mirzadeh et al., 2025). Each experiment uses items not used in any earlier experiment of this series, except one transcoder study (CT-1, Table 1), which reused the items of a resampling experiment (DP2) by design. Earlier draws are reconstructed and excluded before any token is generated. Templates may repeat across experiments. One experiment (DP6) instead uses a synthetic name-lookup task (§3.4).

Experiments. Table 1 lists the experiments. We refer to them by their short names; the results are organised by claim, not by experiment.

Table 1: The experiments, in the order they were run (the closed-model studies of Appendices D and E and the Japanese check of Appendix F are not listed).

Name	Model	Items run / analysed	What it measures	Status
DP1	OLMo-2 1B	100 / 100	forced-swap forks (greedy continuation)	pre-registered
DP2	OLMo-2 1B	60 / 60	resampling forks over 8-token windows	pre-registered
DP3	OLMo-2 1B	40 / 39	both definitions at the same positions; digits	pre-registered

Name	Model	Items run / analysed	What it measures	Status
CT-1	Gemma-2-2b-it	DP2's items (56 pairs)	transcoder explanation of forks (Appendix C)	pre-registered
DP4	OLMo-2 1B	40 / 40	both definitions, stratified drop	pre-registered
DP5	Gemma-2-2b-it	40 / 39	replication on a second model; digit source	pre-registered
DP6	Gemma-2-2b-it	40 / 40	name lookup instead of arithmetic	pre-registered
R1	—	records of DP3–DP6	weight-only baseline, swap size, readout, robustness	post hoc (frozen plan)
DP7	Gemma-2-2b-it	40 / 39	probability-matched forced swap	pre-registered
DP8-X	—	records of DP4, DP5, DP7	effects of natural and forced alternatives by rarity	post hoc (views fixed in advance)
DP8	Gemma-2-2b-it	40 / 38	natural versus forced alternatives by rarity	pre-registered
DP9	Qwen2.5-7B-Instruct	40 / 39	replication of DP5 and DP8 on a larger model of a third family	pre-registered

Base path. Greedy decoding (ties to the lower token id), at most 512 tokens. The answer is taken from the final-answer line, falling back to the last number (DP6: a name read from the final-answer line); numbers are compared exactly. Items whose base answer cannot be read are excluded. We analyse positions before the first answer token a (the token that contains the first character of the answer): in DP3–DP9, every 4th (DP3), 6th (DP4) or 8th (DP5–DP9) token from the start of the response. A *digit position* is one whose base-path token contains a digit.

Fork quantities (Figure 1). At position t, let w_1 be the top token (with probability p_1) and w_2 the runner-up (p_2). We fix the token at t and sample K continuations at temperature 1.0 ($K = 32$ in DP3–DP9). Let q be the fraction of continuations whose answer differs from the base answer; an unreadable answer counts as different.

- *Forced-swap effect*: $e(t) = q_{\text{swap}}(t) - q_{\text{keep}}(t)$, with the token at t fixed to w_2 versus w_1.
- *Probability-matched swap effect (DP7)*: $e_\tau(t) = q_{\text{swap}\tau}(t) - q_{\text{keep}}(t)$, with the token at t fixed to w_τ, the token whose probability at t is closest to $\tau = 10^{-6}$ in log space (w_1, end-of-sequence and special tokens excluded; w_2 allowed; ties to the lower id). A position is *matched* if $p(w_\tau)$ lies within ± 0.5 decades of τ.
- *Forced fork (DP1, DP3)*: the single greedy continuation after the swap changes the answer.
- *Resampling drop*: $d(t) = q_{\text{free}}(t) - q_{\text{keep}}(t)$, where "free" also samples the token at t. d is how much the chance of a changed answer falls once the token at t is fixed to the greedy choice, that is, how much of the outcome is decided at t. DP2 uses a window version, $q(t) - q(t + 8)$, with 16 samples.

Stratified drop. Sampling the token at t from the model's own distribution gives exactly

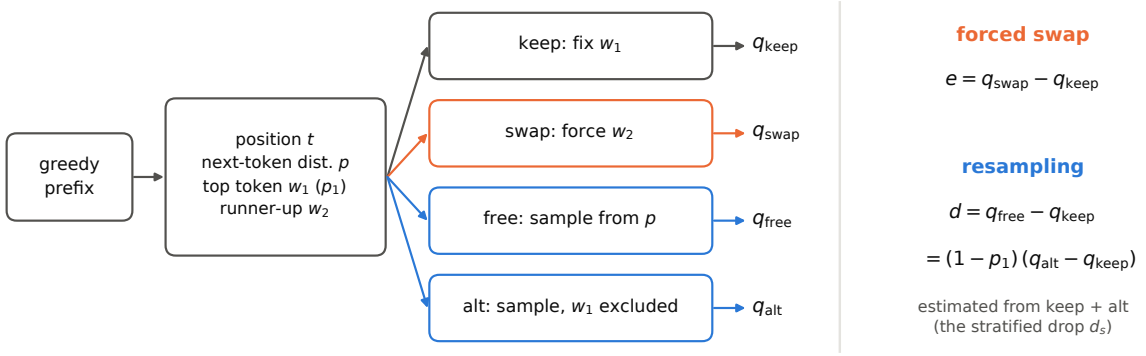
$$q_{\text{free}} = p_1\, q_{\text{keep}} + (1 - p_1)\, q_{\text{alt}}, \qquad \text{so} \qquad d = (1 - p_1)\,(q_{\text{alt}} - q_{\text{keep}}),$$

where q_{alt} samples the token at t with w_1 removed and the rest renormalised. The identity holds for the underlying probabilities, up to numerical differences between the logits used for p_1 and those used for sampling (Appendix A).

- $d_s = (1 - p_1) \cdot e_{\text{alt}}$, with $e_{\text{alt}} = q_{\text{alt}} - q_{\text{keep}}$ (the effect of replacing w_1 by a sampled alternative), is unbiased for d.
- p_1 is read off the logits rather than sampled, so the variance of d_s is $(1 - p_1)^2$ times that of the estimated e_{alt}. Its standard deviation shrinks in proportion to $1 - p_1$ at confident positions; that of the plain estimate does not.
- It needs the same number of samples as the plain estimate (keep + alt instead of keep + free).

Conditioning on the first token and reweighting follows Forking Paths (Bigelow et al., 2025), whose outcome distribution is a mixture over alternative tokens weighted by their probabilities; d_s is its two-group case (the greedy token versus all others). Using the exact probability of the greedy category instead of sampling which category occurs is a form of Rao–Blackwellization (cf. Liu et al., 2019). More generally, Re et al. (2026) use the next-token distributions available during sampling to reduce the variance of expectation estimates

under language models; the stratified drop applies this idea at a single position with two strata. The identity itself is an application of the law of total expectation. At the sentence level, Thought Anchors makes the analogous split between resampling importance and counterfactual importance (Bogdan et al., 2025). We claim no novelty for any of these, only for using the $(1-p_1)$ factor to explain the entropy relation (Appendix A).



q: probability that the final answer differs from the greedy answer, estimated per branch from 32 continuations at T = 1.0.

Figure 1: The two definitions of a forking token. At position t of the greedy path, the token is fixed to the top token w_1 (keep) or to the runner-up w_2 (swap), or it is sampled from the model's distribution, either in full (free) or with w_1 excluded (alt). Each q is the probability that the final answer differs from the greedy answer, estimated from 32 continuations at temperature 1.0 (DP3–DP9). The forced-swap effect e compares swap with keep; the resampling drop d compares free with keep, and the identity on the right holds for these probabilities. DP3 ran keep, swap and free and used the plain estimate of d. DP4–DP6 ran keep, swap and alt at every position and rank positions by the stratified drop d_s, estimated from keep and alt; free was run only to check this estimator and its noise (at every position in DP4, at every third position in DP5 and DP6). DP7 ran keep, swap and a third forced swap to a token of probability about 10^{-6} (not shown); DP8 ran keep, swap, alt and that swap at every position (§3.5); DP9 ran the same four at every position and free at every third (§3.6).

Evaluation.

- Labels are the top 15% of all positions in an experiment (pooled across items) by e, e_τ, d (DP3), e_{alt} or d_s; in DP7, among matched positions only. Ties at the cut are broken by item order, then position.
- The score is the next-token entropy at t in nats, computed from the temperature-1.0 distribution that is sampled from (after soft-capping for Gemma).
- We report the within-item AUROC: computed per item that has both classes, then averaged with equal weight per item. Items with no labelled position (or only labelled positions) drop out, so each estimate rests on fewer items than were run (for example 36 of 39 for top e in DP5), and a difference between two AUROCs can average over slightly different sets of items.
- CIs are 95% percentile bootstrap intervals with 2,000 resamples of templates, labels held fixed; DP3–DP5, DP7–DP9 drew one item per template, and DP6 resamples items (each has its own roster). Differences use the same resamples. Intervals are per comparison and not adjusted for multiple comparisons.
- DP8 does not label positions. Its tests are position-weighted mean differences, a ratio of sums and a mean within-item Spearman correlation (§3.5), with intervals from the same item bootstrap; items whose base answer cannot be read have no positions and enter only the base accuracy. DP9 repeats DP5's labelled tests and DP8's first three tests with the same definitions, and adds one new test (§3.6).

Pre-registration and audit.

- Each confirmatory experiment (DP1–DP9 and CT-1) had a gate document fixing definitions, seeds, labels and decision thresholds. It was frozen and hashed before running. Appendix B lists every decision rule and its outcome.
- R1 is a set of post hoc analyses of existing records, run from a plan frozen before computing them; we label its results "post hoc". Other exploratory analyses (DP1-X, DP1-X2, DP2-X1, DP2-X2, the DP2 side analyses and DP3-X1, DP8-X; described in Appendix B) are also marked post hoc. Analyses that a gate lists as secondary but without a decision rule are called pre-specified secondary analyses.
- After each confirmatory run, a replay audit regenerated a pre-declared subset of outputs with reference

implementations: base paths, continuations or samples on 5–10 items (in DP7–DP9 also the choice of w_τ, in DP8 and DP9 also the first token of alt, in DP9 also free). Every audit matched (0 diverged items).
- In the main text we state each registered outcome in words (confirmed, not confirmed, not replicated or undecided under its rule). The frozen rules and the registered outcome labels are listed in Appendix B (Table 8). "Label" in the main text always means membership in the top 15% of positions.

Origins. The series began with one definition per experiment on OLMo-2 1B: forced swaps (DP1) and resampling over 8-token windows (DP2). Within items, entropy was anti-aligned with forced-swap forks (post hoc) and moderately aligned with resampling forks, and a post hoc look at DP1's records suggested that weighting by the runner-up's probability nearly reproduced the gap. These observations motivated DP3–DP5 (Appendix G).

3 Results

3.1 Forced-swap effects are anti-aligned with entropy on arithmetic

At the same positions, entropy's within-item AUROC for the top 15% of forced-swap effects lies below 0.5 in every arithmetic experiment (Table 2, Figure 2). Comparing it with 0.5 was a registered test in DP4, DP5, DP7 (where the runner-up swap was the prerequisite test) and DP9, and post hoc in DP3.

Table 2: Entropy's within-item AUROC under each definition, with 95% CIs. Registered outcomes for top e: anti-alignment confirmed in DP4, DP5, DP7 and DP9, not replicated in DP6. For top d_s, alignment was a registered test only in DP9 (confirmed); in DP4–DP6 this value entered only the weighting test.

Experiment	Model, items analysed	Positions	Entropy AUROC, top e	Entropy AUROC, top resampling drop
DP3	OLMo-2 1B, 39	1,692	0.415 [0.353, 0.482]	0.558 [0.495, 0.620] (plain d)
DP4	OLMo-2 1B, 40	1,046	0.378 [0.316, 0.440]	0.887 [0.862, 0.911] (d_s)
DP5	Gemma-2-2b-it, 39	905	0.286 [0.211, 0.370]	0.858 [0.821, 0.891] (d_s)
DP7	Gemma-2-2b-it, 39	811 (matched)	0.203 [0.160, 0.246]	— (not measured)
DP9 (§3.6)	Qwen2.5-7B-Instruct, 39	1,324	0.259 [0.200, 0.321]	0.745 [0.703, 0.786] (d_s)
DP6 (name lookup, §3.4)	Gemma-2-2b-it, 40	388	0.574 [0.488, 0.658]	0.860 [0.807, 0.906] (d_s)

- On arithmetic (DP3–DP5, DP7, DP9) every forced-swap interval lies below 0.5. In the name-lookup check (DP6) the forced-swap side is indistinguishable from 0.5, while the resampling side is far above it.
- The anti-alignment was first separated from chance in DP3 (post hoc: 0.415 against a within-item permutation null of mean 0.500, two-sided p = 0.0025); DP3's registered control is discussed in Appendix B.

Robustness (post hoc, R1, and pre-specified secondary analyses of DP7).

- *Threshold.* Labelling the top 5, 10, 20, 30 or 50% instead of 15% leaves every arithmetic point estimate (DP3–DP5, DP7) below 0.5, and the anti-alignment roughly grows as the cut tightens (DP5: 0.194 [0.115, 0.295] at 5% and 0.403 [0.343, 0.465] at 50%; DP7, top e_τ: 0.177 [0.105, 0.264] and 0.427 [0.375, 0.479]). The only interval that reaches 0.5 is DP3's at 50% ([0.418, 0.504]).
- *No labels.* The mean within-item Spearman correlation between entropy and e is −0.092 [−0.148, −0.038] (DP3), −0.149 [−0.226, −0.067] (DP4) and −0.214 [−0.278, −0.150] (DP5), all outside within-item permutation nulls centred near 0; in DP7 it is −0.268 for e and −0.195 for e_τ (permutation 95% ranges about ±0.08).
- *Ties and unreadable answers.* Breaking ties at the cut at random, or counting unreadable continuations as unchanged or dropping them instead of counting them as changed, moves the AUROCs by about 0.03 at most; for the answer-format variants every arithmetic interval stays below 0.5. Unreadable answers are rare (0–3.5% of continuations).
- *Reliability.* The labels are stable: in DP7 the split-half correlation of e is 0.854 and of e_τ 0.876.

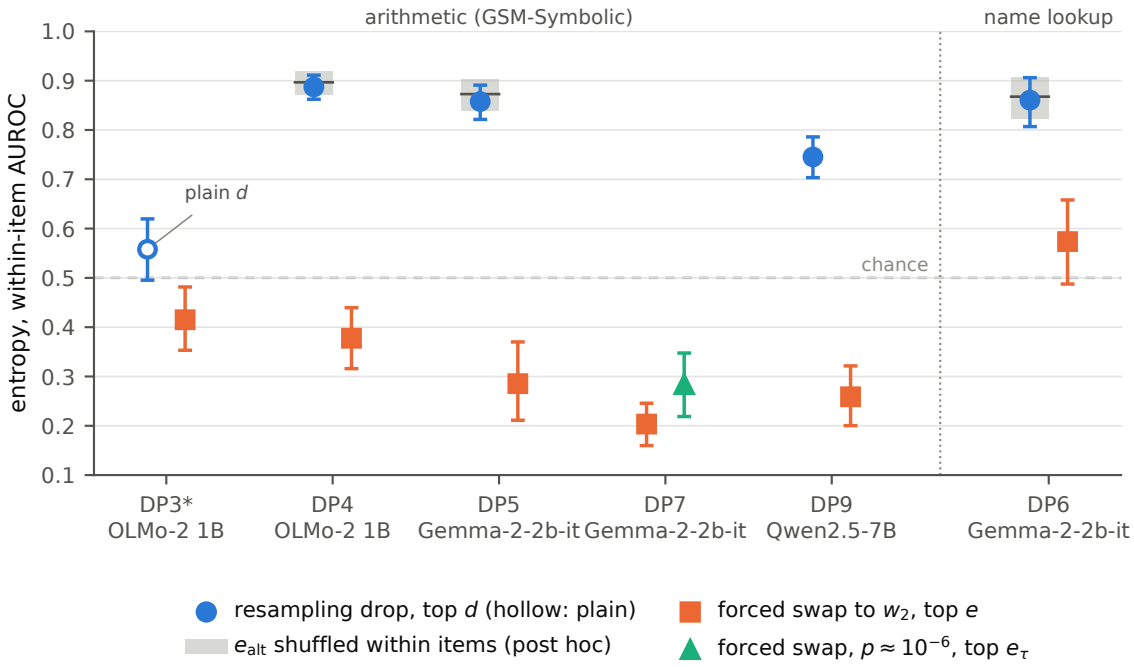


Figure 2: Entropy's within-item AUROC for the top 15% of positions under each definition (Table 2). *DP3: comparing either side with 0.5 was post hoc; DP3 is where the anti-alignment was first separated from chance. Blue circles: resampling drop (hollow: plain estimate of d, DP3; filled: stratified d_s). Grey bands: the same AUROC when each item's e_{alt} is shuffled across its positions while $1 - p_1$ is kept (post hoc, R1; band = 95% range of 2,000 shuffles, line = mean; R1 predates DP9, so DP9 has no band). Orange squares: forced swap to the runner-up (top e). Green triangle: forced swap to a token of probability about 10^{-6} (top e_τ, DP7, matched positions). DP9 is Qwen2.5-7B-Instruct (§3.6). Vertical bars are 95% bootstrap intervals (template-level; item-level in DP6); the dashed line is chance.

- *Headroom (post hoc, HR-1, plan frozen before computing).* A forced swap can raise the chance of a changed answer by at most $1 - q_{\text{keep}}$, so if q_{keep} rose with entropy within items, headroom alone could produce anti-alignment. It did not: within items, entropy and q_{keep} were essentially uncorrelated (mean Spearman between −0.011 and +0.068, every interval including 0; DP4, DP5, DP7, DP9). Dividing e by $1 - q_{\text{keep}}$, or keeping only positions with $q_{\text{keep}} \leq 0.5$, left every interval below 0.5 (DP4 0.348 and 0.335, DP5 0.234 and 0.280, DP7 0.189 and 0.149, DP9 0.260 and 0.238; upper ends 0.418 at most), so the anti-alignment is not a headroom effect.

Pre-specified secondary analyses: solved versus unsolved items. The anti-alignment is strongest on items the model solves. Entropy's AUROC for top e on solved versus unsolved items is 0.343 [0.211, 0.497] versus 0.456 [0.401, 0.508] (DP3), 0.320 [0.238, 0.396] versus 0.442 [0.353, 0.539] (DP4), 0.151 [0.098, 0.214] versus 0.420 [0.303, 0.547] (DP5), 0.139 [0.098, 0.183] versus 0.289 [0.224, 0.346] (DP7) and 0.247 [0.180, 0.318] versus 0.337 [0.286, 0.399] (DP9, where only 5 items were unsolved). On unsolved items the interval includes 0.5 in DP3 (only just), DP4 and DP5, but not in DP7 or DP9. One reason for the gap is mechanical: $e \leq 1 - q_{\text{keep}}$, and q_{keep} tends to be high on unsolved items (on DP1's records, mean 0.64 versus 0.15 on solved items; post hoc).

3.2 The resampling side's alignment largely follows from the definition

The weighting test (pre-registered). By the identity, $d_s = (1 - p_1) \cdot e_{\text{alt}}$. The weighting tests compare entropy's AUROC for top d_s with that for top e_{alt} at the same positions (Table 3, Figure 3).

Table 3: The weighting tests (pre-registered) and the weight-only baseline (post hoc; R1 predates DP9).

	DP4 OLMo-2 1B	DP5 Gemma-2-2b-it	DP9 Qwen2.5-7B-Instruct	DP6 Gemma-2-2b-it (name lookup)
Entropy AUROC, top d_s	0.887	0.858	0.745	0.860
Entropy AUROC, top e_{alt}	0.352	0.301	0.239	0.505
Difference	+0.535 [0.465, 0.604]	+0.557 [0.465, 0.649]	+0.506 [0.438, 0.570]	+0.356 [0.260, 0.447]
Registered outcome	confirmed	confirmed	confirmed	confirmed
Estimator check, mean($d_s - d_{\text{plain}}$)	0.00001 [−0.0055, +0.0056]	−0.0027 [−0.0121, +0.0071]	−0.0002 [−0.0039, +0.0037]	−0.0013 [−0.0138, +0.0112]
Weight-only baseline, mean [95% range] (post hoc, R1)	0.897 [0.872, 0.920]	0.873 [0.840, 0.904]	not computed	0.868 [0.823, 0.906]

Multiplying by $(1-p_1)$ turns an anti-aligned quantity (e_{alt}) into a strongly aligned one (d_s). In DP3, which used the plain estimator, the difference (top d minus top e) was +0.143 [0.035, 0.245], in the predicted direction but short of the registered margin of 0.05 (undecided under its rule); its labels were too noisy (see *Measurement*, below).

How much is built in. A positive result was expected in advance. A power simulation written while DP4 was running, before any results were seen (Appendix A), confirmed the weighting in 60 of 60 runs at every assumed relation between entropy and effect size, including none at all, where the top-d_s AUROC was still about 0.85. The reason is that $1 - p_1$ ranges over several orders of magnitude across positions, while e_{alt} lies between −1 and 1, so the ranking by d_s is set mostly by $1 - p_1$. R1 measured this on the real records: keeping each position's $1 - p_1$ and entropy and shuffling e_{alt} within items gives the AUROCs in the last row of Table 3, and the observed values lie inside those ranges in all three experiments where it was computed (in 77%, 83% and 64% of shuffles the shuffled value was at least as high as the observed one). Ranking by $1 - p_1$ alone gives 0.990, 0.998 and 0.967. In DP6, e_{alt} was at chance (0.505), yet d_s reached 0.860. The resampling side's alignment with entropy is therefore indistinguishable from what the weight produces with effects unrelated to position; the shuffles vary by 0.05–0.08, so a small contribution of e_{alt} is not excluded. We read the weighting outcomes as confirming that the decomposition behaves as derived, not as evidence for a mechanism.

Measurement: the stratified drop makes resampling labels usable. With the plain estimator, resampling labels are very noisy at feasible sample sizes: the split-half correlation of the estimated d (16 versus 16 samples) was −0.02 in DP3. At the same positions and sampling cost, stratifying on the first token raised it from −0.02 to 0.43 (DP4) and from 0.04 to 0.43 (DP5), without changing the estimand (the estimator checks in Table 3 stay within ±0.02; pre-specified secondary analyses). The agreement of the top-15% labels between halves rose less (DP4: 0.765 → 0.834; DP5, same positions: 0.783 → 0.847). The plain estimator's standard deviation does not shrink at confident positions, while the true drop is bounded by $1 - p_1$ and is near zero there (Appendix A). In Qwen2.5-7B-Instruct (DP9) d_s was much less reliable (split-half 0.14 over all positions, against 0.68 in DP5), though still above the plain estimate on its free positions (−0.04); in that very confident model d_s is close to zero at most positions (§3.6).

3.3 What drives the anti-alignment: confident digit tokens, not only the size of the swap

Digits (pre-registered).

- DP3 (OLMo-2 1B): forced swaps change the answer at 58% of digit positions versus 29% of others (+0.30 [0.22, 0.39]); entropy's AUROC for forced forks rises from 0.481 to 0.572 without digits (+0.091 [0.037, 0.155]); confirmed.
- DP5 (Gemma-2-2b-it), the first registered test that digits drive the anti-alignment (motivated by a secondary analysis of DP4 without a decision rule, read post hoc: OLMo's top-e AUROC was 0.378 overall and 0.464 without digits). Digit positions are 20% of positions but about 64% of the top-e positions; the rate of top-e positions is 0.473 among digit positions versus 0.067 elsewhere (+0.406

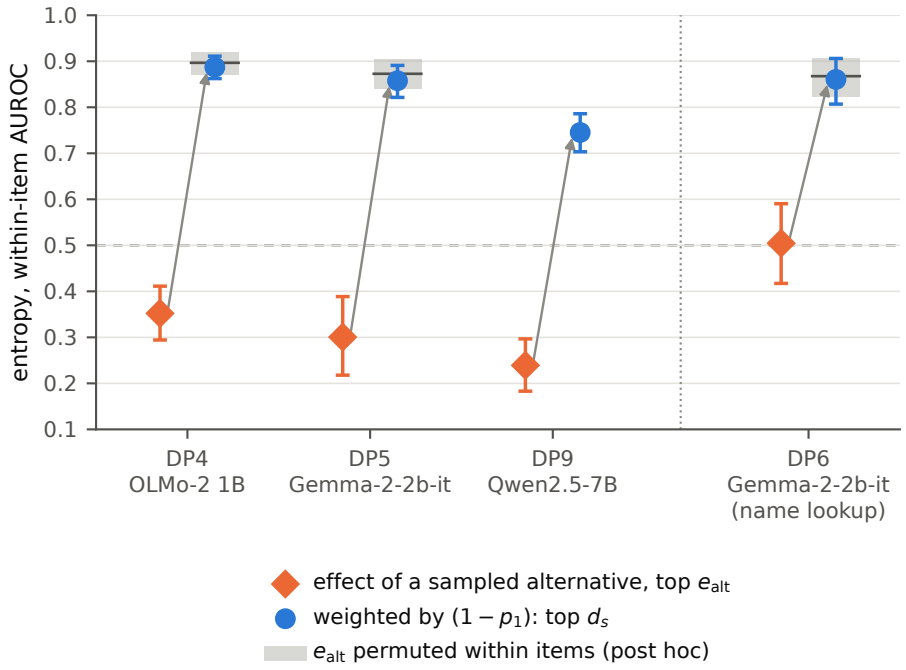


Figure 3: The weighting step, at the same positions. Orange diamonds: entropy's within-item AUROC for the top 15% of positions by $e_{alt} = q_{alt} - q_{keep}$, the effect of replacing w_1 by a sampled alternative. Blue circles: the same for $d_s = (1 - p_1) \cdot e_{alt}$. The arrows mark the multiplication by $(1 - p_1)$. Grey bands: the top-d_s AUROC with e_{alt} shuffled within items and $1 - p_1$ kept (post hoc, R1; 95% range of 2,000 shuffles, line = mean; not computed for DP9, Qwen2.5-7B-Instruct). Bars are 95% bootstrap intervals (template-level; item-level in DP6); the dashed line is chance. In DP6, e_{alt} was not anti-aligned to begin with, so the weighting raised the AUROC from chance rather than reversing it.

[0.314, 0.505]). Removing digits moves entropy's top-e AUROC from 0.286 to 0.452 (+0.166 [0.058, 0.285]); confirmed. Digit tokens are confident: mean entropy 0.08 versus 0.28 elsewhere.

- Without digits, DP5's interval [0.327, 0.583] includes 0.5, and only 17 items have both classes there. Restricted post hoc to those 17 items for both values, the difference is +0.099 [0.027, 0.177] (R1). We claim that digit tokens carry a substantial part of the anti-alignment, not all of it.
- DP9 (Qwen2.5-7B-Instruct) replicated DP5's test with the same rule. Digit positions are 19% of positions but about 60% of the top-e positions (top-e rate 0.482 versus 0.073; +0.408 [0.331, 0.487]). Removing digits moves the top-e AUROC from 0.259 to 0.485 [0.363, 0.597] (+0.226 [0.133, 0.320], 18 items with both classes); confirmed. Digit tokens are even more confident there: mean entropy 0.016 versus 0.174 elsewhere.

Entropy and swap size are entangled in observational data. Entropy and the runner-up's probability p_2 rank positions almost identically (Spearman 0.985, 0.976, 0.988 and 0.937 in DP3–DP6). "Large forced-swap effects sit at low-entropy positions" is therefore also "they sit where the runner-up is improbable", where a forced swap is a large, off-policy perturbation (cf. Macar et al., 2026). Two post hoc analyses (R1) bear on this without separating it:

- Holding p_2 fixed (strata of item × p_2 quintile), digit positions are still much more likely to be top e: Mantel–Haenszel odds ratio 5.12 [3.26, 8.65] (DP3), 6.15 [3.54, 12.45] (DP4) and 23.19 [10.50, 112.04] (DP5); a label-permutation control was centred at 1 in each (medians 0.97–1.01).
- Among non-digit positions, the share of top-e positions barely changes across p_2 tertiles (DP3 13/9/12%, DP4 14/9/9%, DP5 7/6/7%, from the most to the least confident third; descriptive). At digit positions, effects are largest when the runner-up is another digit (top-e rate 48–58%) and small when it is whitespace (0–9%).

The probability-matched swap (DP7, pre-registered). To separate confidence from swap size, DP7 forced, at every position, a second swap to the token w_τ whose probability is closest to $\tau = 10^{-6}$, so that the swapped-in token is about equally improbable everywhere. 95.3% of positions were matched (811 of 851; $\log_{10} p(w_\tau)$ between −6.20 and −5.81 for 90% of them). If the anti-alignment came from swapping in more improbable tokens at confident positions, it should vanish for e_τ; if it came from the positions themselves,

it should remain (Table 4).

Table 4: The pre-registered tests of the probability-matched swap (full rules in Appendix B).

DP7 test (matched positions)	Value [95% CI]	Rule	Outcome
Entropy AUROC, top e (runner-up swap; prerequisite)	0.203 [0.160, 0.246]	upper $<$ 0.50	confirmed (prerequisite met)
Entropy AUROC, top e_τ (probability-matched swap)	0.283 [0.219, 0.348]	upper $<$ 0.50	confirmed (persists)
Odds ratio of top e_τ for digit positions, strata item $\times$ entropy quintile	48.8 [22.9, 231.7]	lower > 1	confirmed (digits more fragile)

- The anti-alignment persists when the swapped-in token's probability is held fixed. Large effects of a matched swap still sit at confident positions (Figure 4a): the mean e_τ is 0.260 in the most confident entropy tertile and 0.075 and 0.072 in the other two.
- At equal entropy, digit positions are far more fragile. They are 19% of matched positions and 72% of the top-e_τ positions (top-e_τ rate 0.572 versus 0.052). The label-permutation control was centred at 1.04.
- Swap size is not irrelevant. In the most uncertain tertile, e_τ exceeds e by 0.056 on average (an improbable token does break uncertain positions more often than the runner-up), and the AUROC for top e_τ (0.283) is somewhat closer to 0.5 than that for top e (0.203). This difference was not a registered comparison and has no interval; the two intervals overlap.
- What is swapped in still differs between positions. At digit positions, 96 of 152 matched tokens w_τ were themselves digit-bearing (Figure 4b), so a matched swap there often replaces one number by another. Inserting a token that is neither digit-bearing nor whitespace at a digit position was also fragile (top-e_τ rate 0.385, 39 positions), against 0.055 for such tokens at non-digit positions; we cannot apportion the effect between the position and the kind of token swapped in.
- *Rank of the matched token (post hoc, RK-1, plan frozen before computing).* DP7 fixed the probability of w_τ, not its rank. We recomputed DP7's base paths teacher-forced with DP7's own code. The greedy token, w_τ and its probability (within 1%) were reproduced at all 6,651 steps and 851 analysed positions. The rank did vary, but w_τ ranked nearer the top where the model was confident: median rank 4, 12 and 27 from the most to the least confident entropy tertile (within-item Spearman between entropy and log rank 0.83 [0.81, 0.86]), and higher-ranked w_τ went with larger e_τ (Spearman between log rank and e_τ -0.16 [-0.24, -0.07]). So where effects were large, the inserted token was not further from the model's choices by rank. Split at the median rank (11), the anti-alignment holds where w_τ ranks 11 or better (0.299 [0.241, 0.360], 31 items with both classes). Where it ranks lower, it is undecided (0.431 [0.324, 0.536]; only 33 top-e_τ positions in 16 items), so we cannot exclude that rank contributes part of it. Because rank and entropy are so closely linked, that half consists mostly of uncertain positions and covers only part of the entropy range. Holding item $\times$ log-rank quintile fixed, digit positions remain far more fragile (odds ratio 66.8 [33.2, 262.0], one bootstrap resample undefined; permutation control centred at 1.01).
- The 40 unmatched positions were the most confident ones (median entropy 4.5×10^{-5} versus 2.4×10^{-2}), where no token lies near 10^{-6}. Excluding them, as registered, is expected to work against the anti-alignment.
- In Qwen2.5-7B-Instruct this control was made descriptive before the run, because a smoke run had matched only 47% of positions; in the run only 16% of digit positions were matched (§3.6). DP7's conclusion rests on Gemma alone.

Other features (exploratory, EX-1, Appendix H). A sweep of 20 features of a position, frozen before computing and without correction for multiple comparisons, found nothing beyond confidence and digits. The nine features that met a mechanical candidate rule in DP4 and DP5 and then held in DP7 and DP9 all measure confidence or digits. At non-digit positions, no feature pointed the same way across the experiments, although with 16–18 items with both classes in three of them this absence has little power.

Readout (post hoc, R1). Digits in the last computation line feed the final-answer line directly, so part of the fragility could be readout rather than computation (cf. Garcia, 2026). Removing the final-answer line, the last non-empty line before it and every number equal to the answer (labels kept) leaves Gemma's anti-alignment unchanged (DP5: 0.286 $\rightarrow$ 0.274 [0.192, 0.356]; R1 reading "anti-alignment survives readout").

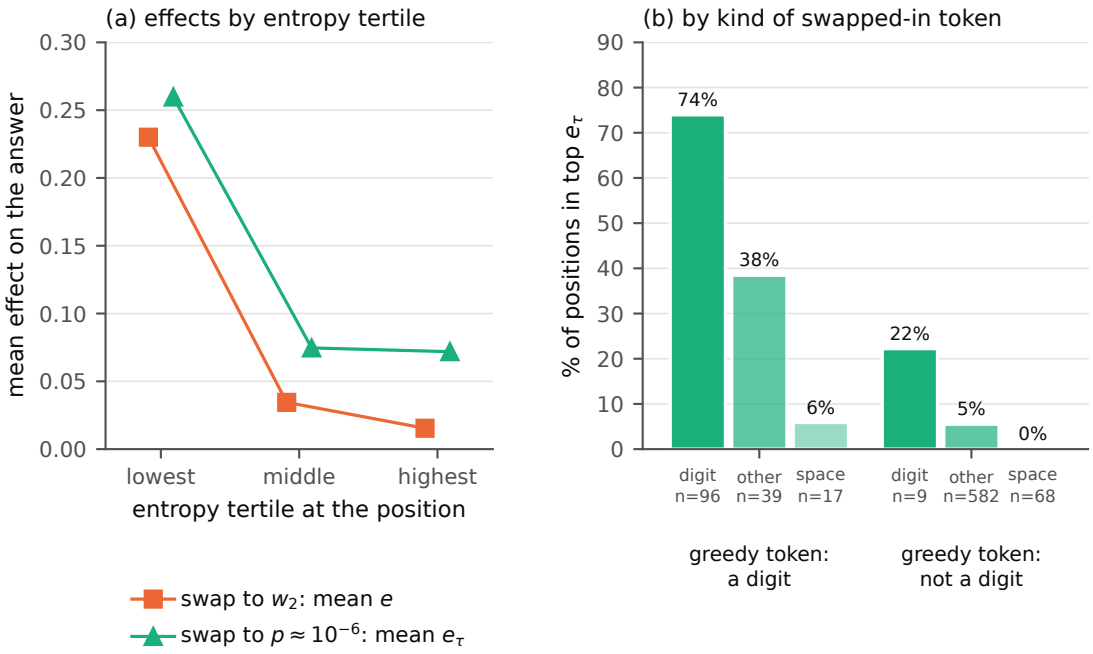


Figure 4: The probability-matched swap (DP7, matched positions; pre-specified secondary analyses). (a) Mean effect of forcing the runner-up (e, orange squares) and of forcing a token of probability about 10^{-6} (e_τ, green triangles), by entropy tertile of the position (270–271 positions each). (b) Share of positions in the top 15% of e_τ, by whether the greedy token is a digit and by the kind of token swapped in (digit-bearing, other, whitespace); n is the number of positions.

For OLMo the result is undecided (DP3: 0.415 → 0.464 [0.386, 0.547]; DP4: 0.378 → 0.433 [0.353, 0.517]): the shifts are within what removing the same number of random positions produces, and most of them come from 8 (DP3) and 16 (DP4) items whose positions all lie in the removed lines and which drop out. Removing only the numbers equal to the answer leaves all three intervals below 0.5.

3.4 A boundary: name lookup (DP6)

Design (pre-registered).

- A synthetic task with name-valued answers (English, fictional names). Each item has its own roster of eight people and twelve relation records. The model identifies a starting person, follows a fixed number of records ("hops") and answers with a full name.
- Name positions (tokens overlapping a roster name) play the role of digit positions.
- The number of hops was set per model by a separate calibration run: 20 fresh items per hop count (one to three), greedy answers only, no fork quantities. The rule picked the hop count with accuracy closest to 0.55 within 0.30–0.80 (else within 0.20–0.90; ties to more hops). The answer-line reading was amended before any model was run.
- Fork quantities, sampling and analysis are as in DP5; intervals use an item-level bootstrap. Table 5 compares the two tasks.

Models run.

- OLMo-2 1B could not perform the task: calibration accuracy 0.15, 0.05 and 0.10 at one to three hops, near chance for eight names and below the pre-registered guard of 0.20. It therefore had no main run.
- Gemma-2-2b-it: calibration accuracy 0.50, 0.85 and 0.45 at one to three hops (we had predicted two hops), so the rule chose one hop. With 20 items per hop count these accuracies have standard errors of about 0.08–0.11, so the choice of one hop is partly chance. Main run: 40 items, 388 positions. Base accuracy 0.725; estimator check −0.0013 [−0.0138, +0.0112]; replay audit: 0 diverged items.

Table 5: Arithmetic (DP5) and name lookup (DP6) compared, both on Gemma-2-2b-it.

	DP5 (arithmetic)	DP6 (name lookup)
Entropy AUROC, top d_s − top e_{alt}	+0.557 [0.465, 0.649]	+0.356 [0.260, 0.447]
Entropy AUROC, top e_{alt}	0.301 [0.218, 0.389]	0.505 [0.417, 0.590]
Entropy AUROC, top e	0.286 [0.211, 0.370]	0.574 [0.488, 0.658]
Registered outcome for top e	anti-alignment confirmed	not replicated
Entropy AUROC, top e, digit / name positions removed	0.452	0.591
Difference (removed − all)	+0.166 [0.058, 0.285] (confirmed)	+0.018 [−0.068, +0.109] (undecided)
Mean entropy, digit / name positions vs others	0.08 vs 0.28	0.13 vs 0.69
Share of top-e positions that are digit / name positions (share of all positions)	64% (20%)	28% (24%)

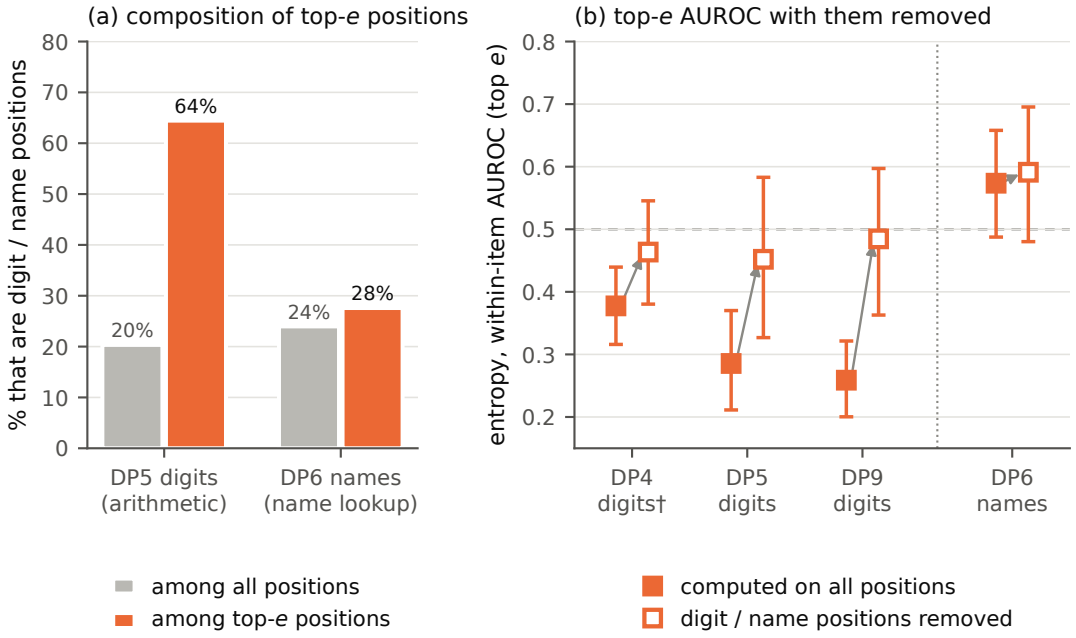


Figure 5: Digit tokens (arithmetic) versus name tokens (name lookup). (a) Grey: share of analysed positions that are digit positions (DP5, 184 of 905) or name positions (DP6, 93 of 388). Orange: their share among the top-e positions (DP5, 87 of 135; DP6, 16 of 58). This re-expresses the registered comparison of top-e rates: 0.473 at digit positions versus 0.067 elsewhere in DP5, and 0.172 at name positions versus 0.142 elsewhere in DP6 (read descriptively). (b) Entropy's within-item AUROC for top e on all positions (filled) and with digit or name positions removed (hollow), with 95% bootstrap intervals (template-level; item-level in DP6); DP9 is Qwen2.5-7B-Instruct (§3.3, §3.6). The top-e labels are those of the full analysis and are not re-ranked after removal. †DP4: a secondary analysis listed in the DP4 gate without a decision rule; reading it as a digit effect came after the run and motivated the DP5 hypothesis (§3.3).

- The weighting replicated, as expected by construction (§3.2). Here nothing flipped: e_{alt} showed no anti-alignment (0.505 [0.417, 0.590]), unlike in DP4 (0.352) and DP5 (0.301).
- The anti-alignment did not replicate. The value lies inside the 95% range of a within-item permutation control (2,000 permutations), [0.404, 0.595]. The label-free rank correlation is also null (0.005 [−0.102, 0.103]; post hoc).
- Labels were noisier than in DP5 (split-half correlation of e 0.56 versus 0.85). This pulls the AUROC towards 0.5, so a weak anti-alignment is not excluded. Post hoc, labelling only the top 5% or 10% gives AUROCs *above* 0.5 (0.637 [0.562, 0.711] and 0.647 [0.565, 0.731]; 15 and 25 items with both classes).

- With no anti-alignment to explain, the registered test of whether name tokens drive it was undecided. Holding p_2 fixed, name positions were not separated from others (odds ratio 2.67 [0.87, 18.28]; post hoc).
- Secondary: even on correctly solved items, no anti-alignment was seen (0.547 [0.426, 0.659]; DP5: 0.151).
- Our predictions were wrong: we had expected anti-alignment (0.35–0.45) driven by name tokens.

Post hoc. Name tokens were confident, like digits, but large forced-swap effects were not noticeably concentrated on them (Figure 5): the rate of top-e positions was 0.17 at name positions versus 0.14 elsewhere (difference +0.030 [−0.037, +0.100]; this is the registered condition of the name test, read here descriptively). One reading, consistent with this but untested, is that the anti-alignment needs tokens that are both confident and answer-breaking when swapped: in arithmetic, digits are such tokens; in this lookup task, names were not disproportionately so. DP6's runner-ups were also less improbable: the most confident third of positions had p_2 below 2.4×10^{-3}, against 5.0×10^{-5} in DP4 and 2.2×10^{-4} in DP5. DP7 argues against a pure swap-size explanation of the arithmetic result, but DP6 itself did not vary swap size.

Scope. One model, one synthetic task, one hop; the change of task is confounded with chain depth. How the two-model gate was applied after OLMo-2 1B dropped out (a choice made after seeing the result) is recorded in Appendix B.

3.5 Rare alternatives: large effects when they occur, little contribution to how answers vary

The forced-swap effects of §3.3 sit at confident positions, where an alternative is rarely sampled. Two questions follow. Is an alternative that the model itself samples as damaging as one that we force? And how much do such positions contribute to how the answer varies under ordinary sampling? We write $r = 1 - p_1$ for the chance that any alternative is sampled at a position.

Exploratory origin (DP8-X, post hoc). We rebuilt the position tables of DP4, DP5 and DP7 from their records with each experiment's own code (each experiment's analysis was reproduced to within 10^{-9} of the recorded result), split positions into six bands of r and compared the effects, with views fixed before looking. Four patterns appeared. In every band e_{alt} and e were about equal. The largest effects sat in the rarest band at digit positions. Positions with $r \leq 0.05$ contributed 4.4% (DP4, OLMo-2 1B) and 3.7% (DP5, Gemma-2-2b-it) of the summed drop d_s. In DP7, the 10^{-6} token broke slightly more often than the runner-up in every band. DP8 tested these four patterns on new items.

Design (DP8, pre-registered). Gemma-2-2b-it, 40 new GSM-Symbolic items (all earlier items and the candidates of the closed-model study of Appendix D excluded), positions every 8th token. At each position four branches of 32 continuations: keep, swap (w_2), alt (the first token sampled with w_1 removed, as in DP4–DP6, with the id of the sampled token recorded) and the probability-matched swap of DP7. Bands of r have edges 10^{-4}, 10^{-3}, 10^{-2}, 0.05 and 0.2. Base accuracy was 0.650; 38 items had a readable base answer (826 positions); 96.6% of positions were matched; the replay audit (5 items × 3 points, all four branches, also the first token of alt and the choice of w_τ) found 0 diverged items; Table 6 gives the tests.

Table 6: The pre-registered tests of DP8 (full rules in Appendix B). All four came out as predicted.

DP8 test	Value [95% CI]	Rule	Outcome
Natural alternative versus runner-up: mean of $e - e_{\text{alt}}$ (all positions)	−0.004 [−0.010, +0.003]	interval within $\pm 0.03 \rightarrow$ like	confirmed (alike)
Rarer alternatives hurt more: within-item Spearman of r and e_{alt}	−0.340 [−0.405, −0.276]	upper < 0	confirmed (rarer hurts more)
Share of the summed drop d_s from positions with $r \leq 0.05$	−0.002 [−0.043, +0.026]	upper < 0.10	confirmed (little)
Probability-matched swap versus runner-up: mean of $e_\tau - e$ (matched positions)	+0.030 [+0.016, +0.044]	lower > 0	confirmed (10^{-6} token hurts more)

- **An alternative the model samples is about as damaging as a forced one, and most damaging where it is rarest.** In the rarest band ($r \leq 10^{-4}$; 225 positions, 56% of them digits) the

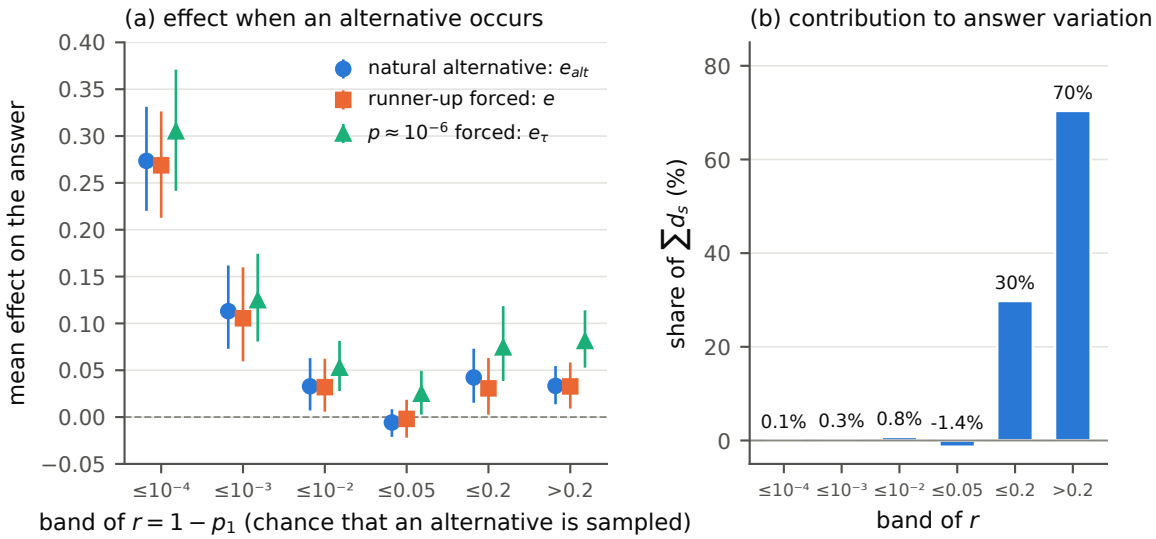


Figure 6: Rare alternatives (DP8, Gemma-2-2b-it, 826 positions). (a) Mean effect on the answer of the three alternatives at a position, by band of r $= 1 - p_1$ (labels give each band's upper edge): an alternative sampled from the model's distribution with the greedy token excluded (e_{alt}, blue circles), the runner-up forced (e, orange squares) and a token of probability about 10^{-6} forced (e_τ, green triangles; matched positions). Bars are 95% bootstrap intervals over items. (b) Share of the summed stratified drop d_s = r·e_{alt} that comes from each band (pre-specified secondary analysis).

three effects were 0.273, 0.269 and 0.305 (e_τ on its 201 matched positions); $e - e_{\text{alt}}$ there was −0.005 [−0.021, +0.012], and at digit positions of all bands −0.013 [−0.034, +0.007] (pre-specified secondary analyses; the latter is not within ±0.03). In the rarest band at digit positions e_{alt} was 0.485 [0.394, 0.588]. At non-digit positions e_{alt} stayed between −0.006 and +0.031 in every band (pre-specified secondary analyses).

- **Not only the runner-up.** The natural alternative was the runner-up in 56% of draws in the rarest band and in 65–79% in the others. Counting only draws whose first token was not w_2, e_{alt} was 0.304 [0.247, 0.366] in the rarest band (digits 0.530) and 0.119 in the next one: alternatives other than the runner-up were about as damaging. We had expected the rarest band's draws to be mostly the runner-up; they were a majority, but a smaller one than in any other band.
- **Rare positions contribute almost nothing to how answers vary.** Positions with r > 0.2 carried 70% of the summed drop and those with $0.05 <$ r ≤ 0.2 another 30%; the four bands with r ≤ 0.05 together −0.2% (Figure 6b). A large effect at a position where an alternative almost never occurs barely moves the answer distribution. The identity bounds each position's drop ($|d_s| \leq$ r), but the bound alone does not make the share small: the 113 positions of the band $0.01 <$ r ≤ 0.05 alone could together carry up to their Σr, which is at least 1.13 (113 × 0.01), against a total of 1.61.
- **Swap size matters a little.** The 10^{-6} token changed the answer slightly more often than the runner-up (+0.030); at non-digit positions the difference was positive in every band and its interval excluded 0 in three of the six; unreadable answers rose with it (up to 4.8% of continuations).

Read together with §3.1–§3.3 (our interpretation; DP8 itself has no entropy test): the fragility of confident digits does not appear to be an artefact of forcing, since the model's own departures from such tokens were at least about as damaging; but those departures are so rare that nearly all of the variation in sampled answers arises at positions where alternatives are common. This is consistent with reading the anti-alignment of forced-swap effects with entropy and the alignment of the resampling drop with entropy as two views of the same positions: the largest effects sit where alternatives are least likely. DP9 repeated DP8's first three tests in a larger model (§3.6).

3.6 A larger model of a third family (DP9)

Every result in §3.1–§3.5 came from models of 1.5B to 2.6B parameters. DP9 asks whether they hold in a model about three to five times larger, from another family.

Design (pre-registered replication). Qwen2.5-7B-Instruct (7.6B parameters), 40 new GSM-Symbolic items, positions every 8th token, and at each position the four branches of DP8 (keep, swap, alt, the

probability-matched swap), 32 continuations each, plus free at every third position for the estimator check. The questions, chosen after all earlier results had been seen, repeat DP5's three tests and DP8's first three with their original rules, and add one: whether entropy's AUROC for top d_s exceeds 0.5 (in DP4–DP6 this value entered only the weighting test). Two choices were fixed before freezing, after a smoke run on four hand-written problems:

- *The probability-matched swap is described, not tested.* Only 47% of the smoke run's positions had a token within ± 0.5 decades of 10^{-6} (Gemma: 95.3% in DP7, 96.6% in DP8); the unmatched ones were almost all at the most confident positions, so a test on matched positions would compare a biased half.
- *An information guard replaces the accuracy range* (a 7B model could exceed 0.90): outcomes would be withheld if too few positions or items carried information, and an unreadable test would be reported as "not confirmed", never as "not replicated" (thresholds in Appendix B).

Run. Base accuracy 0.85; 39 items had a readable base answer (1,324 positions, 247 of them digit positions). No guard was triggered, the estimator check passed (Table 3), and the replay audit (5 items × 3 points, all branches drawn at each point, the first tokens of alt and free and the choice of w_τ) found 0 diverged items. Every rule and value is in Appendix B (Table 8, DP9 rows).

- **Six of seven tests came out as predicted.** The anti-alignment (0.259 [0.200, 0.321], Table 2), the weighting (+0.506 [0.438, 0.570], Table 3), the digit concentration (+0.226 [0.133, 0.320], §3.3), the equivalence of natural and forced alternatives ($e - e_{\text{alt}}$ +0.001 [−0.004, +0.006]) and the rarity correlation (−0.319 [−0.382, −0.256]) replicated, in the same direction and of similar size as in the smaller models. A within-item permutation control (2,000 permutations) gave top d_s a mean of 0.500 (95% range [0.435, 0.568]) and top e 0.501 ([0.442, 0.562]). On the 34 solved and the 5 unsolved items separately, the anti-alignment, the alignment with the drop and the rarity correlation kept their direction, and $e - e_{\text{alt}}$ stayed within ± 0.03 (pre-specified secondary analyses).
- **The alignment with the resampling drop was confirmed but weakly measured** (0.745 [0.703, 0.786], lower than 0.86–0.89 in the smaller models). The model is far more confident: 59% of positions had $r \leq 10^{-4}$ (DP8: 27%), so d_s is near zero at most positions. Its split-half correlation was only 0.14, and the top-15% cut for d_s was 2.9×10^{-7}, so top d_s comes close to separating positions with any sampled drop from those with none.
- **The contribution of rare positions was undecided:** 5.4% [−0.9%, +16.9%] against a 10% line (the gate had predicted "little"). This is not evidence that rare positions contribute; the test could not separate the share from 10%. Positions with $r > 0.05$ carried 94.6% of Σ d_s (68% from $r > 0.2$), as in Gemma; most of the rest came from the band $0.01 < r \leq 0.05$ (4.1%), where Gemma's share was about zero.
- **Rarity, as in DP8.** The large effects sat in the rarest band at digit positions (e_{alt} 0.300, against 0.485 in DP8), and non-digit positions stayed between 0.009 and 0.028 in every band. In the rarest band, where 65% of alt draws began with the runner-up, the draws that did not gave e_{alt} 0.100, against 0.097 for all draws.
- **The probability-matched swap (description only).** 54.5% of positions were matched: 16% of digit positions, 29% of the rarest band and 82–98% of the other bands. On matched positions entropy's AUROC was 0.493 for top e_τ and 0.486 [0.403, 0.571] for top e (labels ranked among matched positions): the anti-alignment is not detectable in this half even for the runner-up swap, so the half says nothing about DP7's question. Our reading is that the matched half consists mostly of non-digit positions (the non-digit AUROC of the digit test was 0.485). On the same positions, $e_\tau - e$ was +0.006 [−0.001, +0.014] (DP8: +0.030), and the digit odds ratio of DP7's third test was 3.77 [1.27, 13.6] (DP7: 48.8; label permutation control centred at 0.95).
- **Format.** 3.6–11.8% of continuations in the keep, swap, alt and probability-matched branches reached the 512-token limit without ending (most often at $r > 0.2$), against under 0.1% in DP8, but answers were readable in more than 99.3% of continuations in every branch and band.

4 Discussion

"Is entropy a good marker of forking tokens?" has no answer until "forking" is defined.

- A forced swap finds positions where *a change would matter*. In our arithmetic chains of thought these are dominated by answer-bearing digits, where the model is confident. Holding the probability of the swapped-in token fixed did not remove the pattern, so it reflects the positions (and what can be swapped in there), not only how improbable the forced token is. In the name-lookup check, the

name tokens, also confident, did not dominate the large effects, and the anti-alignment did not appear. Which tasks produce it is open.

- Resampling finds positions where *a change is likely and would matter*. Its weighting factor $(1 - p_1)$ moves with entropy, and a weight-only baseline reproduces entropy's alignment with the drop. Any drop is bounded by $1 - p_1$, so in this setting entropy marks large drops well almost regardless of where the effects are large.
- The two notions meet at rare alternatives (§3.5). Where the model is confident, the alternatives it would sample itself were about as damaging as a forced runner-up, so the forced-swap picture does not appear to be an artefact of forcing. This rests mainly on the all-positions test; at digit positions alone the difference (-0.013 [-0.034, $+0.007$]) was not shown to lie within ± 0.03, while in the rarest band it was (-0.005 [-0.021, $+0.012$]). But such alternatives almost never occur, so these positions contribute almost nothing to how the answer varies under sampling. A position can be pivotal in the counterfactual sense and irrelevant in the sampling sense.
- The picture is not specific to the two small models: the anti-alignment, the weighting, the digit concentration and the first two rarity results replicated in one 7B model of a third family (§3.6). That model's confidence limited what could be tested there (the contribution of rare positions, the probability-matched control).

Both definitions are legitimate, but they answer different questions. Positions picked by entropy are close to the second notion largely by construction. We have not tested this in a training setting. Studies of forced interventions should report answer-bearing tokens separately, and studies of resampling should compare against a weight-only baseline.

Outside the claims (Appendices D–F). We also pre-registered an attempt to measure resampling forks from outside a closed model whose token probabilities are not exposed (Claude Haiku 4.5 through its API). A first run stopped before resampling, because too few baseline paths met the length condition. An amendment ran in full (99 GSM-Symbolic items, 226,704 resamples, audit passed) but stopped at its pre-registered entry conditions: only 1.4% of words had a positive stratified drop (5% required) and only 19 items had both classes (30 required), so no test of its stand-in for entropy was issued (Appendix D). A further series on that model found forks by observed divergence and confirmed them by forcing; alternatives seen only once or twice in 48 samples did not change the answer more often when forced (mean 0.001 [-0.027, 0.028]); DP8's band of comparable rarity also showed no effect, a correspondence rather than a test, since units, model and data differ. Three hidden forks among words that never diverged in 18 samples were found by sampling more and replicated (Appendix E). A Japanese check with one further open model (Sarashina2.2-1B, 20 MGSM items) showed the weighting result and the anti-alignment (0.409 [0.347, 0.470]), with an undecided digit test (Appendix F).

For measurement within this setting:

- Within-item evaluation avoids rewarding item difficulty.
- Stratifying the resampling drop on the first token makes its estimates markedly more reliable at $K = 32$. The plain estimator's standard deviation (about 0.11–0.12 per position at $K = 32$ in our simulation) does not shrink at confident positions, while the true drop is near zero there. So signal-to-noise is worst there, and that is where stratification removes most of the noise. In a very confident model it stays noisy (§3.6).
- To separate confidence from swap size, fix the probability of the swapped-in token rather than conditioning on the runner-up's probability, which in our data is nearly the same quantity as entropy. In very confident models the target probability may need to be lower, or set per position (§3.6).

5 Related work

Entropy as a fork signal. High-entropy tokens are increasingly treated as the decision points of chain-of-thought reasoning. In reinforcement learning with verifiable rewards, restricting updates to the roughly 20% highest-entropy "forking tokens" matches or beats full updates (Wang et al., 2025); high-entropy regions are linked to pivotal and self-correcting steps (Cheng et al., 2026); and token-level credit has been focused on high-entropy "critical decision pivots" (Yu et al., 2026). Policy-entropy collapse has also been identified as a limit on RL gains (Cui et al., 2025). At inference time, entropy decides where to branch (Li et al., 2026) and where to intervene (Yang et al., 2026), and entropy jumps decide where to cut and resample (Zhou et al., 2026). CoT decoding instead branches on top-k alternatives at the first decoding step and uses

the answer's probability margin as a confidence signal (Wang & Zhou, 2024). Apart from a coarse check by Zhou et al. (resampling at the largest entropy jumps yields more distinct answers than at the smallest), these works do not compare entropy, at the same token positions, with forks defined by intervention.

Doubts about entropy-based selection. For CoT compression, selecting steps by entropy is no better than random pruning, and the apparent token-level benefit of keeping low-entropy tokens on mathematics comes from numeric tokens being low-entropy (Candussio et al., 2026). This partly anticipates our digit mechanism. In the same compression setting, token entropy mixes scaffolding, routing and content, and after an answer-leakage audit no token-level scorer beats a random contiguous block (Papamichalis & Ruane, 2026). When fork points are chosen at step boundaries, shifts in the model's elicited answer belief track Monte Carlo value curves better than entropy, structural and LLM-judge baselines (Lei et al., 2026). Closest to our resampling side, Forking Paths reports that token probability is not strongly predictive of where the outcome distribution changes (a minimal, slightly positive correlation; Bigelow et al., 2025; GPT-3.5, seven tasks). Our within-item analysis of open models (1.5B to 7.6B parameters) on arithmetic instead finds entropy strongly aligned with the stratified resampling drop, and our weight-only baseline attributes that alignment to the $(1 - p_1)$ factor. The settings differ in model, task, signal (entropy versus the probability of the generated token) and evaluation (change-point probability along a text versus a within-item ranking of drops), and we do not resolve the difference.

Forks defined by intervention. Forced edits identify critical tokens by swapping in alternatives and regenerating (Ruan et al., 2025). Rollout-based definitions find tokens after which the chance of a correct answer collapses (Lin et al., 2025) or drops sharply (Ko et al., 2026), or changes sharply in either direction (pivotal tokens; Abdin et al., 2024); Ko et al. classify such "cliff" tokens by greedy choice and token entropy. Forking Paths forces each probable alternative token, resamples continuations after it and weights the outcomes by the token's probability, so it contains both a forced effect per alternative and their probability-weighted mixture; it studies how the outcome distribution moves along a text (Bigelow et al., 2025); Forking Fast estimates these dynamics with fewer samples (Bigelow et al., 2026), and steering is most effective at tokens where the model has not yet committed to an answer (Zur et al., 2025). At the sentence level, Thought Anchors measures the importance of reasoning steps by resampling (Bogdan et al., 2025), and Thought Branches argues that reasoning must be interpreted by on-policy resampling rather than by hand-written, off-policy edits (Macar et al., 2026). Interventions on the chain of thought (early answering, inserted mistakes, paraphrase) were used earlier to measure its faithfulness (Lanham et al., 2023). More recently, Garcia (2026) showed that corruption effects in arithmetic chains can be dominated by the readout of an explicit final-answer line.

What is new here. Forking Paths already contains both quantities, and its mixture contains our stratified estimator. What we add is to measure a forced-swap effect and a resampling drop at the same token positions, in the same items, to compare entropy's relation to both within items, and to do so in pre-registered experiments with replay audits. That forced interventions and resampling disagree has been shown at the sentence level: forced answering highlights sentences doing active computation, while resampling highlights planning and uncertainty management (Bogdan et al., 2025; see also Macar et al., 2026). We show the token-level counterpart: on arithmetic the disagreement reverses the direction of entropy's relation to forks. One side of the reversal follows from the identity $d = (1-p_1)(q_{\text{alt}} - q_{\text{keep}})$, as a weight-only baseline confirms; the other is the concentration of forced-swap effects on confident digits, which persists when the probability of the swapped-in token is held fixed. This control fixes one aspect of off-policyness, the probability of the inserted token, across positions. Its rank still varies, but in the direction opposite to the concern: the inserted token ranks nearer the top at the confident positions where effects are large (post hoc, §3.3), although where the inserted token ranked below the median the anti-alignment was undecided. The control therefore partly addresses the concern of Macar et al. (2026) at the token level. This is consistent with He et al. (2026), who formalise that the information a sampled token carries about the reward cannot exceed its entropy. The resampling drop d therefore has to vanish at confident positions (our identity gives $|d| \leq 1-p_1$ directly), whereas a forced swap is not weighted by probability and need not vanish. That LLMs compute (sub-)answers while generating the chain of thought in multi-step arithmetic (Kudo et al., 2026) fits this picture. Our name-lookup check (§3.4) marks a boundary: the weighting held there, but the anti-alignment did not.

6 Limitations

- **Models.** The claims rest on three open models from three families, 1.5B to 7.6B parameters: OLMo-2 1B and Gemma-2-2b-it in most experiments, Qwen2.5-7B-Instruct in one replication (DP9, 39 items).

No open model larger than 7.6B and no reasoning-tuned model was tested. Qwen2.5 was trained on much mathematical data, and GSM-Symbolic's templates derive from GSM8K, so the problem forms may be familiar to it (not checked). DP9 used scaled-dot-product attention (eager in DP5–DP8) and computed the branches of a position in one merged batch, so its numerical rounding differs slightly from separate computation; the sampled continuations are defined in the same way, and the computation was deterministic in pre-run checks. A pre-registered 7B arm of DP1 was stopped before any output was generated, when its timing check exceeded the gate's 48-hour limit; we then decided not to run it (recorded as an amendment). The probability-matched swap (DP7) was tested on Gemma only: in the 7B model it was made descriptive before the run (a smoke run matched 47% of positions; the run 54.5%, and 16% of digit positions). DP9's replication questions were chosen after all earlier results had been seen, and its gate added one new test (top d_s against 0.5).

- **Task.** All experiments except DP6 (DP1–DP5, DP7–DP9, CT-1) use GSM-Symbolic arithmetic. Our only check outside arithmetic (DP6, §3.4) has one model (OLMo-2 1B could not perform the task), one synthetic task and one hop, and its labels were noisier than in DP5. There the anti-alignment did not appear. The digit mechanism, and the reversal it produces, may be specific to arithmetic; DP6 shows only that the reversal was not seen for one model on one synthetic lookup task at one hop.
- **Construction.** The resampling side's alignment with entropy is reproduced by a weight-only baseline (§3.2). It says little about where reasoning branches.
- **Swap size and token kind.** DP7 held the probability of the swapped-in token fixed but not its kind: at digit positions the matched token was often another digit. Nor did it fix the token's rank. Post hoc (RK-1, §3.3), the anti-alignment was clear where the matched token ranked in the top half, and undecided, with few top positions, where it ranked lower. The somewhat weaker anti-alignment under the matched swap (0.283 versus 0.203) was not a registered comparison and has no interval. We did not test whether the anti-alignment persists among non-digit positions alone under the matched swap; the post hoc analyses of earlier runs (R1) suggest it is weak there.
- **Rare alternatives.** DP8 samples a natural alternative only for the first token; its bands with few digit positions (2–6 in each of the three bands with $r > 10^{-2}$) cannot be read for digits. Its contribution test is a ratio of sums in which the rare bands nearly cancel; we read it as "almost none", not as an exact zero. In the 7B model the same test was undecided (5.4% [−0.9%, +16.9%]), and the stratified drop was much less reliable (split-half 0.14), so "rare positions contribute little" is established for Gemma only.
- **Solved items.** The anti-alignment is concentrated on items the model solves (§3.1).
- **Fork criterion.** Forks are defined by changes in the final answer only, not by changes in the reasoning.
- **Temperature.** All continuations were sampled at temperature 1.0; other temperatures were not tested.
- **Readout.** For Gemma, the anti-alignment survives removing the last two lines and the answer-valued numbers; for OLMo-2 1B the post hoc test was undecided (§3.3). Part of OLMo's digit fragility may be readout rather than computation.
- **Sample sizes and multiplicity.** DP1 has 100 items, DP2 60, and DP3–DP9 about 40 each (DP9: 5 unsolved). Some sub-analyses rest on 11–25 items (non-digit positions in DP5; p_2 tertiles; DP6 at tight thresholds). Intervals are not adjusted for multiple comparisons. The 15% threshold, the position spacing and the tie rule were fixed in advance; varying the threshold and the ties was done post hoc for DP3–DP6 (R1) and, for the threshold, as a pre-specified secondary analysis in DP7 (§3.1).
- **Post-hoc origins.** Several analyses that motivated the confirmatory tests were post hoc (DP1-X, DP1-X2, DP2-X1, DP2-X2, the DP2 side analyses, DP3-X1), and the R1, HR-1 and RK-1 analyses are post hoc although their plans were frozen before computing them; so is DP8-X, whose views were fixed before looking. The confirmatory hypotheses of DP3 (from DP2-X2), DP4 (anti-alignment), DP5 (digit source), DP6 (name source), DP7 (swap size, from R1) and DP8 (from DP8-X, which included the same model, in DP5 and DP7, on other items) were formed after seeing earlier data; DP9 repeated earlier tests with their rules and added one test. Confirmatory claims rest only on the outcomes obtained under pre-registration; the R1 results, including the weight-only baseline, are reported as post hoc support. The within-item reading of DP1, the pooled-AUROC diagnosis and the feature sweep EX-1 (Appendix H; no correction for multiple comparisons) are reported as exploratory.
- **Another language.** The Japanese check (Appendix F) has one small model, 20 items, a base accuracy at the guard's lower end (0.20) and no English counterpart; it was cut from 40 items per language after its projected computing cost was seen (before its effects were computed). MGSM's problems are translations of GSM8K test problems, the same source as GSM-Symbolic's templates, so its problems may overlap with those of the main experiments (with other numbers), and the data family was chosen after the DP results were seen. Whether MGSM or GSM8K is in the model's training data was not

checked and cannot be checked.
- **CT-1.** Underpowered (56 pairs), and its placebo control was higher than expected (Appendix C).

7 Records and code availability

The gate documents, amendments, independent reviews, result notes and result files of every experiment in this paper, with a manifest of their SHA-256 values and of the commits cited here, are at https://github.com/chichichi2001/which-forks (CC BY 4.0). They hold numbers, labels, IDs and hashes only; no benchmark problem text or model output. The analysis code will be released there as a self-contained package with tests that reproduce the published numbers from the frozen inputs.

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A The stratified drop

Identity. At position t let p be the model's next-token distribution at temperature 1.0, w_1 the greedy token with probability p_1, and q_v the probability that the final answer differs from the base answer when the token at t is fixed to v and the rest is sampled. Sampling the token at t from p gives $q_{\text{free}} = \sum_v p(v)\, q_v = p_1\, q_{\text{keep}} + \sum_{v \neq w_1} p(v)\, q_v$. By definition of q_{alt} (the token at t drawn from p restricted to $v \neq w_1$ and renormalised by $1 - p_1$), the second term is $(1 - p_1) \cdot q_{\text{alt}}$. Hence $q_{\text{free}} = p_1 \cdot q_{\text{keep}} + (1 - p_1) \cdot q_{\text{alt}}$ and $d = q_{\text{free}} - q_{\text{keep}} = (1 - p_1)(q_{\text{alt}} - q_{\text{keep}})$. This is exact when (a) the distribution used for p_1 is the one sampled from, (b) keep fixes w_1 and then samples by the same rule, and (c) the same answer rule is used throughout. In our runs (a) holds up to numerical differences: the base path computes one step at a time, while continuations are computed in batches, and in bfloat16 the logits can differ slightly (Gemma: in a pre-run check, logits differed by up to about 0.3–0.5 between cached and uncached computation, and the top token agreed at every step; Qwen2.5-7B-Instruct: up to about 0.3–0.44, and the top token agreed in 31 of 32 steps, the one disagreement falling at a near-tie of one bfloat16 step, 0.125). For Qwen the pre-run check was therefore changed, before any benchmark item was run, from "the top token agrees at every

step" to "at least 29 of 32 steps agree and the largest logit difference is at most 1.0". The estimator check, $\text{mean}(d_s - d_{\text{plain}})$, tests the identity in data (§3.2, §3.4, §3.6).

Restricting to the runner-up gives $d = p_2\, e + \sum_{v \notin \{w_1, w_2\}} p(v)\,(q_v - q_{\text{keep}})$; the $p_2 \cdot e$ of Appendix G is the first term. e_{alt} is the probability-weighted average effect of all alternatives, e the effect of w_2 alone.

Weight-only baseline (R1, post hoc). For each item, the values of e_{alt} are permuted across that item's positions while $1 - p_1$ and entropy stay attached to their positions; d_s is recomputed, the top 15% relabelled and entropy's within-item AUROC recomputed. 2,000 permutations per experiment. The baseline keeps each item's distribution of effects and destroys only their link to position.

Variance. With K samples per branch and independent branches, $\text{Var}(d_{\text{plain}}) \approx [q_{\text{free}}(1-q_{\text{free}}) + q_{\text{keep}}(1-q_{\text{keep}})]/K$ and $\text{Var}(d_s) \approx (1-p_1)^2[q_{\text{alt}}(1-q_{\text{alt}}) + q_{\text{keep}}(1-q_{\text{keep}})]/K$, because p_1 is known exactly. In a simulation at $K = 32$ (20,000 replications; $q_{\text{keep}} = 0.3$, $q_{\text{alt}} = 0.6$, or 0.5 in the last row) both estimators are unbiased, and their standard deviations are given in Table 7.

Table 7: Simulated standard deviations of the plain and stratified estimates of the drop.

p_1	true d	SD of d_{plain}	SD of d_s
0.99	0.003	0.114	0.0012
0.90	0.030	0.117	0.012
0.60	0.120	0.119	0.047
0.30	0.140	0.119	0.084

So the standard deviation is about 95× smaller at $p_1 = 0.99$, about 10× smaller at $p_1 = 0.90$ and about 2.5× smaller at $p_1 = 0.60$.

Power simulation for the weighting test (written while DP4 was running, before any results were seen). Synthetic data with 40 items and 15–45 positions per item; 70% of positions nearly certain (mean entropy 0.05) and 30% uncertain (mean entropy 0.8); $p_1 \approx \exp(-0.9 \cdot \text{entropy})$; large effects (0.3–0.8) occur with a probability tied to entropy by a parameter γ ($\gamma < 0$: fewer large effects at uncertain positions), other effects between −0.05 and 0.08; binomial noise from 32 samples per branch; the gate's labelling and analysis code; 60 runs per setting. For γ from −3 to 0, the top-e_{alt} AUROC ranged from 0.355 to 0.493 and the top-d_s AUROC from 0.798 to 0.848, and `WEIGHTING_EXPLAINS_FLIP` was returned in 60 of 60 runs in every setting.

B Pre-registered tests, decision rules and outcomes

All intervals are two-sided 95% percentile bootstrap intervals over templates (items in DP6), 2,000 resamples. "Within-item AUROC" is entropy's AUROC for the labelled positions, computed per item with both classes and averaged. Rules are quoted as frozen; where a gate gives a precondition, it is listed in the rule (Table 8).

Table 8: Every pre-registered decision rule and its outcome for the open-model experiments (JP-1: Appendix F).

Experiment	Question	Rule (as frozen)	Observed [95% CI]	Label
DP1	Does entropy flag forced-swap forks? (pooled AUROC)	lower ≥ 0.75 → strong; upper < 0.65 → weak; else moderate	0.563 [0.524, 0.593]	`DP1_WEAK_MARKER`
DP2 (1)	Does entropy flag resampling fork windows? (max entropy in the window)	lower ≥ 0.70 → marks; upper < 0.60 → weak; else moderate	0.660 [0.587, 0.727]	`ENTROPY_MODERATE`

Experiment	Question	Rule (as frozen)	Observed [95% CI]	Label
DP2 (2)	Does a hidden-state probe add to cheap features? (AUROC difference)	point $\geq +0.03$ and lower $> 0 \rightarrow$ precursor; upper $< +0.03 \rightarrow$ none; else undecided	-0.081 [-0.172, $+0.015$]	NO_PRECURSOR_BEYOND_CHEAP
DP3 (1)	Does weighting explain the flip? (top d − top e, plain d)	lower $> 0.05 \rightarrow$ explains; upper $< 0.05 \rightarrow$ none; else undecided	$+0.143$ [0.035, 0.245]	WEIGHTING_UNDECIDED
DP3 (2)	Do digits drive forced-swap fragility? (forced forks, non-digit − all)	lower > 0.03 and digit−non-digit fork rate lower $> 0 \rightarrow$ drive; upper $< 0.03 \rightarrow$ none; else undecided	$+0.091$ [0.037, 0.155]; rate $+0.299$ [0.217, 0.391]	DIGITS_DRIVE_FORCED_FRAGILITY
CT-1 (1)	Are forks less explained by the transcoder? (pair win rate)	precondition: median replacement score ≥ 0.45 on the instruction-tuned model, else use the base model (0.557); lower $> 0.55 \rightarrow$ less; upper $< 0.45 \rightarrow$ more; else no clear difference	0.464 [0.327, 0.586]	NO_CLEAR_DIFFERENCE
CT-1 (2)	… beyond cheap features? (win rate on residuals)	lower $> 0.55 \rightarrow$ beyond; upper $< 0.55 \rightarrow$ explained; else undecided	0.518 [0.389, 0.638]	BEYOND_ENTROPY_UNDECIDED
DP4 (1)	Weighting with the stratified estimator (top d_s − top e_{alt})	precondition: estimator check CI within ± 0.02; lower $> 0.05 \rightarrow$ explains; upper $< 0.05 \rightarrow$ none; else undecided	$+0.535$ [0.465, 0.604]	WEIGHTING_EXPLAINS_FLIP
DP4 (2)	Is entropy anti-aligned with top e?	upper $< 0.50 \rightarrow$ away; otherwise lower $> 0.45 \rightarrow$ not replicated; else undecided	0.378 [0.316, 0.440]	ENTROPY_AWAY_FROM_SWAP_EFFECT
DP5 (1)	Weighting, second model	as DP4 (1), estimator checked at every third position	$+0.557$ [0.465, 0.649]	WEIGHTING_EXPLAINS_FLIP
DP5 (2)	Anti-alignment, second model	as DP4 (2)	0.286 [0.211, 0.370]	ENTROPY_AWAY_FROM_SWAP_EFFECT
DP5 (3)	Do digits drive the anti-alignment? (top e, non-digit − all)	lower > 0.03 and top-e rate digit − non-digit lower $> 0 \rightarrow$ drive; upper $< 0.03 \rightarrow$ not from digits; else undecided	$+0.166$ [0.058, 0.285]; rate $+0.406$ [0.314, 0.505]	DIGITS_DRIVE_AWAY
DP6 (1)	Weighting, name lookup	as DP5 (1)	$+0.356$ [0.260, 0.447]	WEIGHTING_EXPLAINS_FLIP
DP6 (2)	Anti-alignment, name lookup	as DP4 (2)	0.574 [0.488, 0.658]	AWAY_NOT_REPLICATED
DP6 (3)	Do names drive the anti-alignment?	as DP5 (3), names for digits	$+0.018$ [-0.068, $+0.109$]; rate $+0.030$ [-0.037, $+0.100$]	NAME_AWAY_UNDECIDED
DP7 (2)	Prerequisite: anti-alignment with top e on new items (matched positions)	upper $< 0.50 \rightarrow$ replicated; otherwise lower $> 0.45 \rightarrow$ not; else undecided; main 1 read weakly unless replicated	0.203 [0.160, 0.246]	AWAY_REPLICATED
DP7 (1)	Does it persist with a probability-matched swap? (top e_τ)	upper $< 0.50 \rightarrow$ persists; otherwise lower $> 0.45 \rightarrow$ gone; else undecided; read weakly if $>$ 10% of positions unmatched	0.283 [0.219, 0.348]	AWAY_PERSISTS_MATCHED
DP7 (3)	Are digit positions fragile at equal entropy? (Mantel–Haenszel odds ratio, item × entropy quintile)	lower $> 1 \rightarrow$ beyond; upper $< 1 \rightarrow$ below; else not separated; control: median of 200 label permutations outside [0.9, 1.1] $\rightarrow$ not read; read weakly if $>$ 10% of positions unmatched	48.8 [22.9, 231.7]; control 1.04	DIGIT_BEYOND_CONFIDENCE_AND_SWAP

Experiment	Question	Rule (as frozen)	Observed [95% CI]	Label
DP8 (1)	Natural alternative as damaging as the runner-up? (mean $e - e_{\text{alt}}$)	interval within $[-0.03, +0.03] \rightarrow$ like; lower $> 0.03 \rightarrow$ runner-up more; upper $< -0.03 \rightarrow$ natural more; else undecided	-0.004 [-0.010, $+0.003$]	NATURAL_LIKE_SECOND
DP8 (2)	Rarer alternatives more damaging? (within-item Spearman of r and e_{alt})	items with ≥ 4 positions and non-constant r and e_{alt}; upper $< 0 \rightarrow$ more; lower $> 0 \rightarrow$ less; else undecided	-0.340 [-0.405, -0.276]	RARER_ALT_HURTS_MORE
DP8 (3)	Rare positions contribute little? (share of $\Sigma\ d_s$ from $r \leq 0.05$)	upper $< 0.10 \rightarrow$ little; lower $> 0.10 \rightarrow$ contribute; else undecided; not read if the total is ≤ 0 or in $> 5\%$ of resamples	-0.002 [-0.043, $+0.026$]; total 1.61	RARE_POSITIONS_CONTRIBUTE_LITTLE
DP8 (4)	10^{-6} token breaks more than the runner-up? (mean $e_\tau - e$, matched)	lower $> 0 \rightarrow$ more; upper $< 0 \rightarrow$ less; else undecided; read weakly if $> 10\%$ of positions unmatched	$+0.030$ [$+0.016$, $+0.044$]	TAU_HURTS_MORE_THAN_SECOND
DP9 (1)	Anti-alignment, third family, 7B (top e)	precondition for all DP9 tests: information guard (thresholds under Guards and audits, below) and at most one diverged audit item; for tests using top labels (1–3) also top-15% cut > 0; as DP4 (2)	0.259 [0.200, 0.321]	ENTROPY_AWAY_FROM_SWAP_EFFECT
DP9 (2)	Weighting (top d_s − top e_{alt})	as DP5 (1)	$+0.506$ [0.438, 0.570]	WEIGHTING_EXPLAINS_FLIP
DP9 (2b)	Alignment with the drop (top d_s)	estimator check as DP5 (1); lower $> 0.50 \rightarrow$ toward; upper $< 0.50 \rightarrow$ away; else undecided	0.745 [0.703, 0.786]	ENTROPY_TOWARD_RESAMPLING_DROP
DP9 (3)	Digits (top e, non-digit − all)	as DP5 (3)	$+0.226$ [0.133, 0.320]; rate $+0.408$ [0.331, 0.487]	DIGITS_DRIVE_AWAY
DP9 (4)	Natural alternative versus runner-up (mean $e - e_{\text{alt}}$)	as DP8 (1)	$+0.001$ [-0.004, $+0.006$]	NATURAL_LIKE_SECOND
DP9 (5)	Rarer alternatives more damaging? (Spearman of r and e_{alt})	as DP8 (2)	-0.319 [-0.382, -0.256]	RARER_ALT_HURTS_MORE
DP9 (6)	Rare positions contribute little? (share of $\Sigma\ d_s$ from $r \leq 0.05$)	as DP8 (3)	$+0.054$ [-0.009, $+0.169$]; total 1.35	RARE_CONTRIBUTION_UNDECIDED

Guards and audits. DP2–DP8 required a base accuracy between 0.20 and 0.90 (observed 0.375–0.725; DP1 had no guard, accuracy 0.52); CT-1 required at least 48 readable items and 30 items with a pair (59 and 33). DP6's calibration dropped OLMo-2 1B by its guard. DP7 would have read its main tests 1 and 3 weakly if more than 10% of positions were unmatched (4.7% were), and DP8 its main test 4 likewise (3.4% were). DP8's intervals resample the items that have positions (38 of 40), with the percentile end points taken as order statistics, as in DP7. DP9 replaced the accuracy range by the information guard of §3.6 (accuracy at least 0.20, at least 80 positions with $1 - p_1 > 0.05$, at least 20 items with both classes for each of top e, top d_s and top e_{alt}; observed 0.85, 246, and 37/26/37), withheld any test whose top-15% cut was ≤ 0 (all cuts were positive; top d_s 2.9×10^{-7}), and resampled the 39 items that have positions as DP8 did. Every replay audit (5 items × 3 points with four reference samples per branch, or 10 items in DP1) found 0 diverged items; the rule was "at most one (DP1: two) diverged item".

Post hoc analyses. DP1-X (within-item reading of DP1; item-level features), DP1-X2 (digits in DP1),

DP2-X1 (item difficulty), DP2-X2 (weighting and digits on DP1's records), the DP2 side analyses (probe power, DP1 records), DP3-X1 (permutation null) and R1 (weight-only baseline, odds ratios given p_2, readout exclusion, thresholds, rank correlation, ties, answer format, runner-up types; plan frozen before computing), and DP8-X (effects by band of rarity on the records of DP4, DP5 and DP7; views fixed before looking). DP9's description of the probability-matched swap (§3.6) was fixed in its gate, without decision rules.

DP3's control. DP3's pre-registered control, a single within-item permutation of the top-d labels, gave 0.433 [0.367, 0.494]. A post hoc null of 2,000 permutations (DP3-X1; mean 0.501, 95% range [0.444, 0.558]) shows the procedure is unbiased; the registered single draw fell in its lowest 1%, which the planned reading of DP3-X1 treats as a rare chance draw.

DP6's gate and one model. The DP6 gate, written for two models, said that if neither showed name tokens driving the anti-alignment, we would report the digit mechanism as possibly specific to arithmetic. It did not foresee one model having no main run. A memo written after calibration, before the main run, had mapped this situation to the gate's case "only one model shows it" (state the result for that model only). After seeing the result we applied the case "neither shows it" instead, because no model was labelled `NAMES_DRIVE_AWAY`.

Closed-model study. The pre-registered study in Claude Haiku 4.5 issued no test labels (CR-1 stopped at G1, CR-1a at its entry conditions). It and the replacement-type analysis on its data (PARA-1a; summarised there, with its rules and all labels in `notes/para1a_partA_result.md`) are reported in Appendix D; the later series CR-2 to CR-5, with its labels, in Appendix E.

Japanese check. The per-language tests of JP-1 (Sarashina2.2-1B, Japanese MGSM; rules as DP5 (1)–(3)), their labels, and why its cross-language tests were not run are reported in Appendix F.

C A negative result for a transcoder explanation of forks (CT-1)

Gemma-2-2b-it with Gemma Scope transcoders (Lieberum et al., 2024; Dunefsky et al., 2024), via the circuit-tracer library (Hanna et al., 2025; Ameisen et al., 2025), on DP2's items (59 readable). Resampling forks were recomputed for Gemma with DP2's window definition (drop ≥ 0.25), giving 56 matched pairs of a forking and an ordinary token from 33 items. The transcoders were trained on the base Gemma 2 2B and are applied here to the instruction-tuned model, a choice fixed in the CT-1 gate.

- Forking and matched ordinary tokens leave the same unexplained fraction (0.434 versus 0.436).
- The win rate (how often the forking token is the less well explained of its pair) is 0.464 [0.327, 0.586], labelled `NO_CLEAR_DIFFERENCE`. After removing what cheap features predict, it is 0.518 [0.389, 0.638] (`BEYOND_ENTROPY_UNDECIDED`).
- We obtained 56 pairs where about 110 were planned, so the intervals are about ± 0.13 wide and a moderate difference is not excluded.
- A placebo control was higher than expected (0.70 [0.50, 0.90], 20 pairs), which we report as a limitation.

D A closed model — resampling forks from outside (CR-1, CR-1a) and the types of its replacements (PARA-1a)

This appendix reports a separately pre-registered study and its outcome, as its gate required in advance: whatever the outcome, in this appendix and in the repository's result notes. It is not part of the paper's claims. All numbers are from the result notes; no item text or model output is shown anywhere.

Design (CR-1, frozen before any request). Model `claude-haiku-4-5-20251001` without extended thinking, asked to think briefly and end with `#### <number>`; GSM-Symbolic, one item per template (100 templates, items of the earlier experiments excluded, up to three candidates per template in a fixed random order).

- *Baseline path*: temperature 0, at most 400 tokens; usable if the answer is readable, the stop reason is `end_turn`, and the text before the answer line has 30–80 units.
- *Units and cuts*: tokens are not visible, so the text before the answer line is split into units by the regular expression `\s*\S+` (leading whitespace belongs to the next unit). Cut c (c = 0, …, L) prefills the first c units as an assistant turn.

- *Resampling*: $K = 48$ continuations per cut with default sampling, of which 12 (group A) measure spread and 36 (group B) measure forks. q(c) is the share of group B whose final answer differs from the baseline's (unreadable answers count as different).
- *Forks*: a group-B continuation from cut w − 1 "keeps" unit w if it starts with s_w (prefix match). With n_B the number of returned group-B continuations (36 when none is missing), m(w) the share that does not keep w, x_diff(w) the number of those whose answer changed, and $q_{\text{keep}}(\text{w}) = \text{q(w)}$ estimated by prefilling unit w, the stratified drop is $d_s(\text{w}) = \text{x_diff(w)/n_B} - \text{m(w)}\cdot q_{\text{keep}}(\text{w})$. This follows Appendix A, except that the kept stratum is estimated by the prefill and the weight m is estimated from the same draws rather than known. Relative forks are the top 15% of all words by d_s, among words with $d_s > 0$, ties included.
- *Stand-in for entropy*: u(w) = 1 − the share of the most common first unit among the 12 group-A continuations from cut w − 1.
- *Tests*: (0) a signal check — continuations that do not keep w change the answer more often than the prefill (the n_diff-weighted mean of g(w) = x_diff/n_diff − q_{keep}, item-bootstrap lower bound > 0); (1) the within-item AUROC of u for relative forks, against a permutation null that exchanges the "answer changed" flags between keeping and diverging continuations within each cut; (2) a baseline that permutes g within items while keeping m (cf. the weight-only baseline of §3.2); (3) a Mantel–Haenszel odds ratio for digit units. Labels for (1)–(3) are issued only if (0) passes, at least 5% of words have $d_s > 0$, and at least 30 items have both relative forks and other words.

CR-1: stopped at G1. Only 57 templates gave a usable baseline path (70 required), so the study stopped before any resampling (`STOPPED_BEFORE_MAIN_RUN`; cost about $0.20). All 134 length failures were too long (median 104 units).

CR-1a: amendment and outcome. The only change was to add "using at most 60 words" to the prompt and to rerun G1 from the same candidates. *Disclosure*: the amendment was designed after seeing CR-1's aggregate G1 numbers (counts, lengths, baseline accuracy); no resampling request had been sent, and no quantity of the main analysis existed. G1 then gave N = 99 items (107 candidates called; baseline accuracy 0.970; mean length 46.7 units; 4,723 cuts). All 226,704 resampling requests succeeded (no retries; every stop reason `end_turn`); 18 of 170,028 group-B answers were unreadable (0.011%). The audit passed: recomputation of 5% of the cuts (237) by a separate reference implementation found no mismatch, and the permutation null was reproducible.

Table 9: CR-1a's entry conditions. Label: `CR-1a: INSUFFICIENT_FORK_SIGNAL`; tests (1)–(3) were not run.

Entry condition	Required	Observed	
(0) weighted mean of g	lower bound > 0	0.00297 [0.00097, 0.00571]	passed
words with $d_s > 0$	≥ 5%	1.36% (63 of 4,624)	**not met**
items with both relative forks and other words	≥ 30	19	**not met**

Further reported numbers: all 63 words with $d_s > 0$ were selected (693 allowed); 41% of them had $d_s \leq 1/36$; 55% of words had no diverging continuation in group B. The two halves of group B (even and odd draws) gave a split-half reliability of d_s of 0.787 and label agreement = 0.441. The total cost of CR-1 and CR-1a was about $53.60, of which resampling (batch API) was $53.32.

Reading (post hoc). Nothing failed in the measurement; the answer simply changed rarely. With a baseline accuracy of 0.97 and chains of about 47 units, resampling from almost any word boundary returned the same answer. The prompt, the model and the item selection all differ from the open-model experiments, so we do not compare the rate of forks with theirs.

PARA-1a: types of replacement in the same data (pre-registered; summarised here, full report in the repository notes `notes/para1a_partA_result.md` and `notes/para1a_explore_ws_result.md`). For each resampled continuation whose first unit differed from the baseline unit, the unit was typed (orthographic variant, empty, digit, inflection, function word, related in meaning (WordNet 3.0; Miller, 1995; via NLTK, Bird et al., 2009) or in spelling (edit distance), other) and its rate compared with a shifted baseline (the baseline unit moved to other cuts of the same item). Of 48,585 continuations from 99 items, as item means, 74.7% were orthographic variants of the baseline unit and 0.87% were related in meaning. Related-in-meaning replacements fell below the shifted baseline (−1.40 points [−2.17, −0.73];

PARA-1a:SEMANTIC_BELOW_MISMATCH_BASELINE), but without orthographic variants (post hoc, plan written before computing) the difference was +0.65 [−0.69, +1.95], consistent with the label reflecting that the replacement was usually the same word, not that the model avoids related words. Whether the type relates to the answer changing was not shown (digit minus semantic −0.02 points [−1.20, +0.82]; PARA-1a:NO_DIFFERENCE_SHOWN); for every type about 1% of continuations changed the answer. The original PARA-1 Part A, tied to CR-1's data, was not run; the amendment that moved PARA-1 to CR-1a's data was written after CR-1's G1 counts and frozen before CR-1a's G1 rerun and before any resampling request.

Whitespace (post hoc, plan written before computing). 72% of the "diverging" continuations differed from the baseline unit only in leading whitespace, which CR-1's prefix rule counts as not keeping it. Counting them as keeping it lowered the share of words with $d_s > 0$ to 1.04% and the number of eligible items to 18, so whitespace was not the main reason CR-1a's signal was weak.

E A closed model, continued — searching for forks, forcing alternatives, and rare alternatives (CR-2 to CR-5)

This appendix reports four further pre-registered experiments on the closed model of Appendix D. Like Appendix D, it is not part of the paper's claims; it is reported because the series bears on §3.5. Each experiment had its own frozen plan, an independent review before freezing, record checks before every paid stage and a reference-implementation audit (all passed). All numbers are from the result notes; no item text, model output or word is shown anywhere, to the authors included.

Setting. `claude-haiku-4-5-20251001` through the batch API (at most 1,024 tokens; default sampling; the baseline path at temperature 0). Data: 90 items of CHASE-Math (Patel et al., 2025; depths 2–5; MIT licence as displayed on the dataset card), which was harder for this model than the GSM-Symbolic items of Appendix D (baseline accuracy 0.66 on a 48-item look, against 0.970 there). The data family was chosen after a look at 48 items of it, and the procedure was designed after an exploration on another family that was then not used; the items themselves were unseen. Units are whitespace-delimited words of the baseline path. For the word at position w, measured by resampling from the cut before it:

- m: the share of continuations that do not write the baseline word (an alternative occurs);
- g: how much more often the final answer changes when an alternative occurs; d_s = m·g (cf. Appendix D);
- u: the spread of the first unit across resamples (a stand-in for entropy, as in Appendix D);
- g_F: the effect of *forcing* the most common alternative word a1 by prefill, q_A1 − q_O.

CR-2: search, then measure ($19.81). A binary search over five points of each path located the window (at most 6 words) where the chance of a changed answer drops most; all window words and two comparison words per item (604 words) were then measured with new samples (6 + 18 per cut). Signal check 0.114 [0.079, 0.160]. u separated relative forks (top 15% by d_s) with within-item AUROC 0.757 [0.690, 0.815] over 49 items (BBU_MODERATE), but a negative control that exchanges outcomes within cuts was not excluded (p = 0.148, WITHIN_CUT_EXCHANGEABILITY_NOT_EXCLUDED) and the difference from a shuffled baseline was undecided (+0.032 [−0.033, +0.098]). Post hoc (views fixed in advance): 340 of the 604 words (56%) never diverged in 18 samples and contained no fork; "diverged at least once" alone gave AUROC 0.866; among the 264 words that did diverge, u was at chance (0.551).

CR-3: force the alternative (the 264 diverging words, $13.90). g_F identified CR-2's forks (within-item AUROC 0.818 [0.741, 0.892], 37 items, p = 0.0005; FORCED_EFFECT_REPRODUCES_FORKS). Within items, neither u nor m showed a rank association with g_F (0.045 [−0.097, 0.179] and 0.069 [−0.075, 0.203]; U_G_ASSOCIATION_NOT_SHOWN, M_G_ASSOCIATION_NOT_SHOWN). Split-half reliability of g_F was 0.79. Forcing a1 was about as damaging as a1 occurring naturally (difference −0.008 [−0.022, 0.006], 174 words; descriptive).

CR-4: hidden forks among words that never diverged (340 words, $5.18). 32 new samples per word found an alternative for 30 words; forcing it (64 samples) identified 3 hidden forks after Holm correction (HIDDEN_FORKS_FOUND; smallest p 3.4×10^{-26}). Whether these words' effects differ from those of the diverging words was undecided (Δ −0.076 [−0.173, +0.021]).

CR-5: replication and a further search ($3.40). With fresh samples only, all 3 hidden forks replicated (FORKS_REPLICATED; g 0.92, 0.45, 0.38) and 2 of 3 words that had passed only a Benjamini–Hochberg correction were confirmed (BH_ONLY_FORKS_CONFIRMED). 16 more samples on the 310 words still without an alternative found one for 11 words, whose alternatives appeared once or twice in 48 samples (mean rate 0.023). Forcing them did not change the answer more often: mean g_F 0.001 [−0.027, 0.028], no word with

g_F $\geq$ 0.25, smallest p 0.237 (`NEW_HIDDEN_FORKS_NOT_DETECTED`). A fork with g = 0.3 would have been detected with probability about 0.8. The 299 words that never diverged in 48 samples were not tested.

Table 10: Effect of forcing the most common alternative, by how rarely it occurred (descriptive; the groups come from different experiments and were not compared by a registered test).

Group of words	How often an alternative occurred	Mean g_F [95% CI]
CR-3: diverged within 18 samples (264)	median 6 of 18	0.119 [0.087, 0.153]
CR-4: first alternative within 32 more (30)	1 to a few of 32	0.090 [0.028, 0.160]
CR-5: first alternative within 48 (11)	1–2 of 48	0.001 [−0.027, 0.028]

Relation to §3.5 (interpretation; units, model and data differ). CR-5's rarest group corresponds to r of about 0.01–0.05, and in that band DP8 also found no effect (mean e_{alt} −0.006, 0 of 113 positions with e_{alt} $\geq$ 0.25). In DP8 the destructive alternatives sat at r $\leq 10^{-3}$, where an alternative appears less than once in 1,000 samples; words of that kind may be among CR-5's 299 untested words and are practically invisible to a search that relies on observed divergence. The two series are therefore consistent with each other (DP8 does not test CR-5): a search by observed divergence finds forks where alternatives are common, and is unlikely to find, at feasible sample sizes, the rare positions where a forced alternative would matter but the model almost never produces one. Total cost of CR-2 to CR-5: $42.29.

F Another language and model family — a Japanese check (JP-1)

This appendix reports a pre-registered check in Japanese. Like Appendices D and E, it is not part of the paper's claims. What ran is a revised, Japanese-only part of the frozen plan, analysed as exploratory. No item text, model output or token is shown anywhere, to the authors included.

Setting.

- Model: `sbintuitions/sarashina2.2-1b-instruct-v0.1` (SB Intuitions, 2025), a Japanese language model according to its card (MIT licence as displayed there).
- Data: the Japanese part of MGSM (Shi et al., 2023), translations of 250 GSM8K test problems. We used the Hugging Face mirror `juletxara/mgsm` at a pinned revision (b2f13d4), whose card displays CC-BY-SA-4.0. Items were chosen by index with a fixed seed.
- Definitions and estimators: as in DP5:
  - positions every s tokens before the answer;
  - $K = 32$ resamples for keep, swap and alt, and free at every third position;
  - the stratified drop d_s;
  - top 15% within the language.
- Prompt and readout: the DP prompt translated into Japanese, with a final line made of the Japanese word for "answer" and the number. The readout rule was fixed with a decision table on harmless example sentences.
- Precision and pre-run check: the model ran in float32. In the first pre-run check, bfloat16 rounding flipped near-ties between the cached greedy path and a full recomputation. The plan was amended before any main-run output (amendment 1), and the repeated check then passed. In both checks the pre-run accuracy on 16 separate items was 8/16 (Japanese) and 6/16 (English), so the 1B model was kept, without switching to the reserve 3B.
- Sampling step: amendment 1 also computed the shared prefix once and copied its cache to the 32 rows. This differs from computing the 32 rows separately in the last digits. In a check, 1–3 of 32 samples diverged within 48 steps. Both sample the same model distribution; they differ only by rounding.
- Audits:
  - Replay audit: it reproduced the saved samples with the same step, so it checks reproducibility, not the sampling distribution. That was checked in the pre-run checks: the cached step against full recomputation, and the GPU sampler against a plain-Python one. It covered 3 items × 3 points, the seeded audit items that fall among the 20. Its rule was made stricter after this audit's result (0 diverged) was known: any diverged item fails, against "two or more" in the frozen plan. 0 items diverged.
  - Reference recomputation of the token categories and the answer token: 1 item, 0 mismatches.

What was planned and what ran.

- The frozen plan had 40 items in each of Japanese and English. Its main questions were:
  - DP5's tests 1–3 in each language;
  - the Japanese − English difference in entropy's within-item AUROC for top e;
  - whether large d_s effects fall on reasoning words or on connectives, with Japanese tokenised by fugashi (McCann, 2020) and UniDic.
- The Japanese first half ran (533 positions, 15.4 hours on one GPU). This half was the items in rows 0–19. The split into halves was fixed by item number in the pre-run lock, before any output.
- The projected total cost (about 150 compute units) then exceeded the plan's stop line (45). The plan was amended (amendment 2), with an independent review and the owner's approval:
  - 20 items per language;
  - for the English half only, a resampling step that drops finished rows from the batch;
  - a compute line of 70 units. Before this amendment we had seen the cost, the resample lengths and end-of-sequence counts, the Japanese audit result and a speed check of the new step (including its sample differences at 4 positions). No effect size or label had been computed.
- The English half stopped at the amended line, at a projected 76.7 units. In the two English items that ran, chains had about 2.4 times as many positions per item as the Japanese mean. In all, JP-1 used about 32 units (about 18.4 GPU-hours).
- After English had stopped, and before any effect was computed, the analysis was restricted to the Japanese half: the frozen rules of DP5's tests 1–3 applied to the 20 items. No cross-language result exists.

Table 11: The Japanese check (JP-1).

Test (rules as DP5 (1)–(3))	Japanese [95% CI] (items with both classes)	Label	DP5 (Gemma-2-2b-it)
Weighting (top d_s − top e_{alt})	+0.411 [0.312, 0.510] (19, 20)	`JA_WEIGHTING_EXPLAINS_FLIP`	+0.557
Anti-alignment (top e)	0.409 [0.347, 0.470] (19)	`JA_ENTROPY_AWAY_FROM_SWAP_EFFECT`	0.286
Digits (top e, non-digit − all; top-e rate digit − non-digit)	−0.013 [−0.084, +0.058] (17 non-digit); rate +0.141 [0.040, 0.247]	`JA_DIGIT_AWAY_UNDECIDED`	+0.166; +0.406

- Values are within-item AUROC; intervals are over items, 2,000 resamples; digits use DP5's token-level marker.
- Of the 20 items, 19 have both classes for top d_s and for top e, and 20 for top e_{alt}.
- No difference between JP-1 and DP5 was tested; items, model and language differ.
- **Guard and checks.**
  - Base accuracy was 0.20 (4 of 20), exactly at the lower end of the guard (0.20–0.90).
  - All 20 answers were readable, all with the answer marker. Under the plan's strict readout the accuracy would be 0.15, and 1 answer matched a known misreading pattern.
  - 79 of the 533 positions are top e. The estimator check passed (mean $d_s - d_{\text{plain}}$ −0.005 [−0.016, +0.004]).
  - Label-permutation controls (labels shuffled within items, 2,000 times) were centred at 0.50 (sd 0.04–0.05).
  - Token categorisation failed for 26 tokens of the base paths (20 empty, 6 prefix mismatches). Only the category summaries use it.
- **Weighting and anti-alignment.** Both appear in this further model and language. The anti-alignment's interval excludes 0.5. Its point estimate is closer to 0.5 than DP5's, but the two were not compared.
- **Digits.** 26% of digit positions are top e, against 12% of other positions. Removing digit positions moved entropy's AUROC from 0.409 to 0.396, a change of −0.013 [−0.084, +0.058] on 17 items. That is undecided, so a weakening is neither shown nor excluded. The point estimate is far from DP5's +0.166, but no test compares them.

- **Solved versus unsolved items** (pre-specified secondary analysis).
  - On the 16 unsolved items the AUROC is 0.415 [0.352, 0.481]. Its interval excludes 0.5, as in DP7 and unlike DP3–DP5 (§3.1).
  - The 4 solved items (3 with both classes) are too few to read.
  - So these results come mostly from unsolved items.

G Where this started — single-definition experiments (DP1, DP2)

The series began with one definition per experiment.

- **Forced swaps (DP1; OLMo-2 1B, 100 items, 18,178 positions; forks are runner-up swaps that change the greedy answer).** Pre-registered: pooled AUROC 0.563 [0.524, 0.593], `DP1_WEAK_MARKER`. Post hoc (DP1-X), within items, entropy is anti-aligned, 0.432 [0.406, 0.459]; on solved items, 0.394 [0.349, 0.439]. Item-level features alone reach a pooled AUROC of 0.74–0.78, so pooled AUROC mixes in item difficulty, and we use within-item AUROC from DP2 on. Forced forks are common: 31.6% of positions overall, 11.8% on solved items and 45.7% on unsolved items.
- **Resampling (DP2; OLMo-2 1B, 60 new items, 1,296 eight-token windows, 16 samples per point).** Real forks (window drop ≥ 0.25) are rare (6.6%). The window's maximum entropy separates them with within-item AUROC 0.660 [0.587, 0.727] (`ENTROPY_MODERATE`). Post hoc, the entropy of the window's first token gives only 0.466 [0.384, 0.550] (DP2-X2). A frozen hidden-state probe did not add to cheap features: -0.081 [-0.172, $+0.015$] (`NO_PRECURSOR_BEYOND_CHEAP`), which we read as "no precursor beyond cheap features with this probe". DP2 is a single estimate with noisy labels; DP3's comparable quantities were lower (0.558 per position; 0.503 [0.33, 0.67] for a one-token analogue of the window, with only 19 positives).
- **The link (post hoc, DP2-X2).** On DP1's resampling records (20 items $\times$ 20 positions, runner-up only), entropy's AUROC is 0.439 [0.376, 0.503] for the top forced-swap effects e and 0.707 [0.654, 0.761] for the top $p_2 \cdot e$: weighting alone nearly reproduced the DP1→DP2 gap. In DP2's windows, forced swaps changed the answer at 62% of digit-initial windows versus 29% of others. These observations motivated DP3–DP5.

H An exploratory sweep of 20 position features (EX-1)

This appendix is exploratory and is not part of the paper's claims. Before computing, we froze a list of 20 features of a position, four views and a mechanical rule for calling a feature a candidate (notes/ex1_feature_sweep_spec.md, 2026-10-18). Apart from entropy, whether the greedy token is a digit, p_1 and p_2, none of the features had been computed in any experiment.

- **Data.** The records of DP4 (OLMo-2 1B), DP5 (Gemma-2-2b-it), DP7 (Gemma, new items) and DP9 (Qwen2.5-7B-Instruct) were re-read at each experiment's run commit and matched the recorded results (exactly, or within 10^{-9}). The distribution features were recomputed with one teacher-forced pass of the same model over the base path. The recomputed greedy token matched the recorded one at 99.3–100% of positions, and the recomputed entropy had a Spearman correlation of at least 0.9997 with the recorded one. DP7's rank feature comes from RK-1 (§3.3).
- **Token kinds** are categories only: digit-bearing, operator ($+ - \times \div * / =$; an ASCII hyphen was also counted as an operator), whitespace or newline, punctuation, word. "Inside a number" means that the previous token is digit-bearing too.
- **Views.** V1: within-item AUROC of the feature for each experiment's top e (DP7: the runner-up swap at matched positions). V2: V1 restricted to non-digit positions, labels kept. Also described, not used for candidates: top d_s (DP4, DP5, DP9) and top e_τ (DP7, DP9).
- **Rule.** A feature is a candidate if, in both DP4 and DP5 and in the same view (V1 or V2), its interval excludes 0.5, the direction agrees and $|\text{AUROC} - 0.5| \geq 0.10$. A candidate replicates within the exploration if DP7's and DP9's intervals in that view exclude 0.5 in the same direction. There is no correction for multiple comparisons, which is why nothing here is a claim.

Result. Nine features were candidates, all in V1, and all nine replicated:

- lower at top-e positions: entropy, Rényi-2 entropy, the number of tokens with p $\geq 10^{-6}$, the rank of the token nearest 10^{-6}, the runner-up's share of the rest, and "the greedy token is a word";

- higher: p_1, the logit gap and "the runner-up is digit-bearing".

Every one of them measures confidence or digits, so they restate the main finding from other angles. The replication adds little for the confidence features, whose relation to top e in DP7 and DP9 was already known, and because all experiments were computed in one run, the spec could not guarantee that DP7 and DP9 were unseen when the candidates were fixed; the rule was applied mechanically for that reason. In V2 (non-digit positions) no feature was a candidate. Only 6 of the 80 cells had intervals excluding 0.5, and none was consistent across experiments; with 16–18 items with both classes in DP5, DP7 and DP9, this absence has little power. Five more features pointed the same way in all four experiments in V1, with intervals excluding 0.5, but fell short of the effect threshold in DP4 or DP5: the position in the chain (relative position and tokens to the answer; later positions were more often top e; 0.583 and 0.417 in DP4), "the runner-up is of the greedy token's kind" (0.567 and 0.575 in DP4 and DP5), and two more confidence features (the probability of the top 5 tokens, 0.581 in DP4; the number of tokens with p $\geq 10^{-3}$, 0.432 in DP4). None of them was seen in V2. "Inside a number" was 0.69–0.72 in DP5, DP7 and DP9 but 0.501 in DP4 (OLMo); we did not check why. Several token-kind and line features had intervals excluding 0.5 with $|\text{AUROC} - 0.5|$ no larger than 0.061. Within this list, then, we found no second axis besides confidence and digits. Under top d_s, the confidence features were aligned (entropy 0.887, 0.858 and 0.745), as the identity of §3.2 implies.

Table 12: Exploratory sweep, V1: within-item AUROC of each feature for the top 15% of e (DP7: runner-up swap, matched positions). Items with both classes: 38, 36, 35, 37 (DP4, DP5, DP7, DP9). * marks a 95% interval (item bootstrap, 2,000 resamples) that excludes 0.5; the intervals are in notes/ex1_result.md.

Feature	DP4	DP5	DP7	DP9
Entropy	0.378*	0.286*	0.203*	0.259*
Top-1 probability p_1	0.631*	0.718*	0.795*	0.742*
Logit gap, top 1 − top 2	0.639*	0.734*	0.802*	0.744*
Probability of the top 5 tokens	0.581*	0.625*	0.733*	0.707*
Number of tokens with p $\geq 10^{-3}$	0.432*	0.362*	0.279*	0.399*
Number of tokens with p $\geq 10^{-6}$	0.395*	0.356*	0.246*	0.305*
Rényi-2 entropy	0.369*	0.283*	0.205*	0.258*
Runner-up share of the rest, $p_2/(1-p_1)$	0.324*	0.244*	0.303*	0.338*
$\log_{10}$ rank of the token nearest 10^{-6}	0.396*	0.353*	0.250*	0.324*
Greedy token is an operator	0.485	0.460*	0.467*	0.474*
Greedy token is whitespace or a newline	0.500	0.460*	0.443*	0.449*
Greedy token is punctuation	0.466*	0.473	0.453*	0.471*
Greedy token is a word	0.367*	0.301*	0.277*	0.294*
Runner-up is digit-bearing	0.690*	0.767*	0.816*	0.796*
Runner-up is of the greedy token's kind	0.567*	0.575*	0.622*	0.566*
Relative position t/a	0.583*	0.711*	0.605*	0.620*
Tokens to the answer	0.417*	0.289*	0.395*	0.380*
Position starts a line	0.500	0.477*	0.482*	0.479*
Inside a number	0.501	0.698*	0.690*	0.720*
Position follows an operator	0.531	0.473	0.440*	0.461*
Greedy token is digit-bearing (reference)	0.682*	0.805*	0.861*	0.812*

Table 13: Exploratory sweep, V2: as Table 12, at non-digit positions only (labels kept); the digit reference row is constant (0.500) by construction and shown as — . Items with both classes: 31, 17, 16, 18 (DP4, DP5, DP7, DP9). * marks a 95% interval (item bootstrap, 2,000 resamples) that excludes 0.5; the intervals are in notes/ex1_result.md.

Feature	DP4	DP5	DP7	DP9
Entropy	0.464	0.452	0.550	0.485
Top-1 probability p_1	0.544	0.551	0.448	0.519
Logit gap, top 1 − top 2	0.545	0.564	0.438	0.509

Feature	DP4	DP5	DP7	DP9
Probability of the top 5 tokens	0.537	0.516	0.432	0.538
Number of tokens with p $\geq 10^{-3}$	0.497	0.468	0.534	0.510
Number of tokens with p $\geq 10^{-6}$	0.473	0.506	0.584	0.493
Rényi-2 entropy	0.456	0.452	0.555	0.481
Runner-up share of the rest, $p_2/(1-p_1)$	0.422*	0.438	0.486	0.453
$\log_{10}$ rank of the token nearest 10^{-6}	0.471	0.505	0.580	0.499
Greedy token is an operator	0.508	0.490	0.524	0.500
Greedy token is whitespace or a newline	0.544	0.538	0.513	0.497
Greedy token is punctuation	0.502	0.584*	0.476	0.551
Greedy token is a word	0.445	0.388*	0.487	0.452
Runner-up is digit-bearing	0.512	0.555	0.476*	0.506
Runner-up is of the greedy token's kind	0.481	0.451	0.479	0.459
Relative position t/a	0.534	0.610	0.436	0.532
Tokens to the answer	0.466	0.390	0.564	0.468
Position starts a line	0.506	0.494	0.486*	0.498
Inside a number	0.500	0.500	0.500	0.500
Position follows an operator	0.590*	0.571	0.464	0.496
Greedy token is digit-bearing (reference)	—	—	—	—